Theory of Computation.

Learning Objectives.

CLO1 • Introduce core concepts in automata & TOC

CLO2 • Identify different formal language classes & their relationship

CLO3 • Learn concepts of Grammers & Recognisers for different formal languages

CLO4 : Prove or disprove Theorems in automata theory

CLO5 : Theorems in automata using properties

Cource Outcomes

CO1 : Understand the fundamental concepts of automata theory & TOC

CO2 : Build DFA, NDFA, ε-NFA, Convert one automata model to another, minimize DFA & Illustrate the properties of regular Languages

CO3 : Develop Context free Grammer & push down automata (PDA) for formal languages & interpret the properties of Context free Languages

CO4 : Construct Turing to solve Computational problems

Module 1

Automata: Any computation can be represented as theoretical frameworks that represents Design of programs to solve a wide variety of problems

Automata is abstract machine which represents Conceptual framework to solve a problem

Eg: Design of Digital circuits, Compiler construction, pattern matching, Software Design, AI video Games, DNA matching etc..

ALPHABET: Is defined as finite, non-empty set of elements denoted by Sigma(Σ) which contains any symbols or characters Eg: *, +, -, /. A to Z etc

STRING: The sequence of symbols obtained from alphabets of a Language

NOTE: The empty string is denoted by Epsilon (ε)

The set of all possible strings over an alphabet Sigma (Σ) is Sigma star (Σ^*) → Kleen closure

* → Zero or more
+ → 1 or more
. → Single place holder
? → 0 or 1

$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \ldots \Sigma^n$ → Kleen star

LANGUAGE: Consists of various set of alphabets from which word or statement can be obtained denoted by (L)

Q) A Language which contains Empty string

→ $L = \{\varepsilon\}$

Q) A Language of strings which contains Equal no of a's & b's

→ $\Sigma = \{a, b\}$

$L = \{\varepsilon, ab, aabb, abab, baba, bbaa, \ldots \infty\}$

Q) Language of strings consist of n no of 0's followed by n's of 1's

Soln

$$\Sigma = \{a, b\}$$

$$L = \{\varepsilon, ab, aabb, aaabbb \ldots\ldots \infty\}$$

Q) Language contains string of 0's & 1's having exactly 1 zero

Soln $\Sigma = \{0, 1\}$

$$L = \{0, 01, 10, \ldots, 011, 10111 \ldots\ldots \infty\}$$

Functions on Strings

1) Length of a string: denoted by modulus

Eg: $w = "apple"$ $|w| = 5$

2) Concatenation of 2 string:

$u = "abc"$ $v = "efg"$

$uv = "abcefg"$

3) Replication of string: denoted by w^i i is a natural number

Eg: $w = 'm'$

$w^0 = \varepsilon$ $w^1 = m$ $w^2 = mm$

Eg: $a^0b^3 = \varepsilon bbb$

4) String reversal denoted by w^R

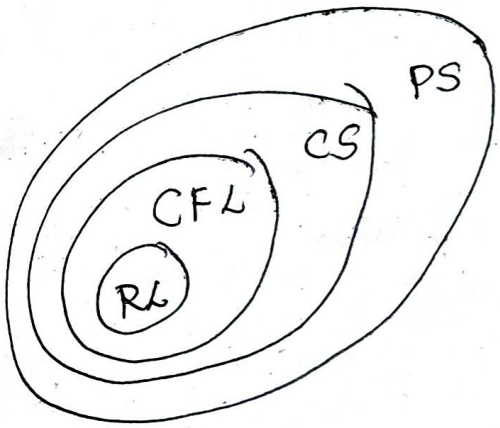
$x = a_1 a_2 \ldots a_n$

$x^R = a_n a_{n-1} \ldots a_2 a_1$

Relations on String

1) Substrings
   a) substring
   b) proper substring
   c) prefix
   d) proper prefix
   e) suffix
   f) proper suffix

Language Hierarchy



1. Type 0 - phrase structured langu
2. Type 1 - Context sensitive
3. Type 2 - Context Free
4. Type 3 - Regular language
   Regular Gramar

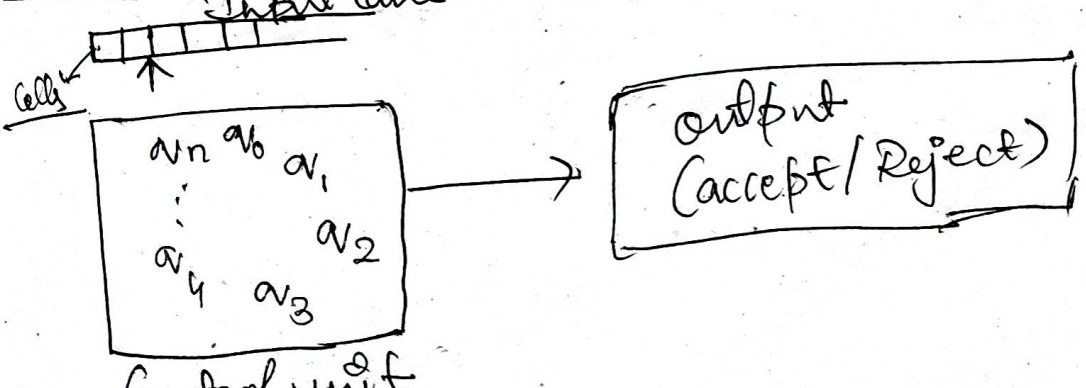
FSM - Finite State Machines or Finite Automata (FA)

→ The simplest & most efficient computational device

→ Finite automata is a mathematical model used to study abstract machine or abstract computing devices with the inputs chosen from Σ

→ Abstract machine is a conceptual model of a computer hardware or Software Systems

General Representation of Finite Automata



DFSM → Deterministic finite State machine
or DFA

→ It is computational device whose input is string and output is accept or reject

FORMAL DEFINITION OF DFA / DFSA / DFSM

$M = (Q, \Sigma, \delta, q_0, F)$ → quin tuples or 5 tuples

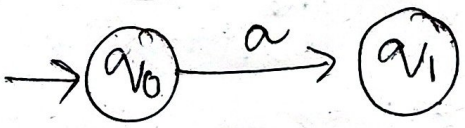
Q = Finite set of states
Σ = Input alphabets
q_0 = Start state
F = Finite state
δ = Transition Function ie it maps from $Q \times \Sigma$ to Q

Symols used In Finite Automata

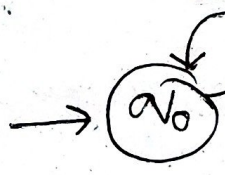
simple state

Start state

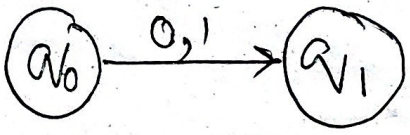
Final/accepted state



$\delta(q_0, a) = q_1$
$Q \times \Sigma \quad Q$



$\delta(q_0, a) = q_0$



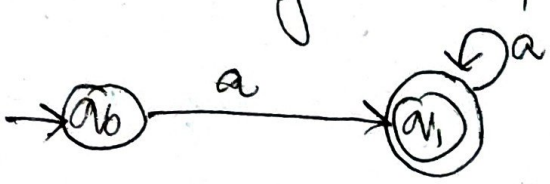
$\delta(q_0, 0) = (q_1)$
$\delta(q_0, 1) = (q_1)$

a) ... at least 1 a,

$\Sigma = \{a\}$

$L = \{a, aa, aaa, \ldots\}$ = a^+

a^+ - Regular Expression



State Transition Diagram

$Q = \{q_0, q_1\}$ State: min string + 1

$\Sigma = \{a\}$

q_0 - start

$F = q_1$

Transitive Table

δ	a
$\rightarrow q_0$	q_1
final state $\{q_1$ or $(q_1)\}$	q_1

a) Any no of a's $\Sigma = \{\varepsilon, a, aa, aaa \ldots\} = a^*$

$Q = \{q_0\}$ $\Sigma = \{a\}$

q_0 - start $F = q_0$

δ	a
$\rightarrow q_0$ or (q_0)	q_0

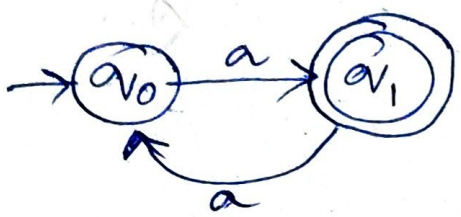
Transition Table

$\delta(q_0, a) = q_0$

Q) Design a DFA with odd no of a's

Sol

(1) → min string $L = \{a, aaa, aaaaa \ldots\}$



$Q = \{q_0, q_1\}$

q_0 - start

$\Sigma = \{a, aaa, aaaaa \ldots\}$

$F = q_1$

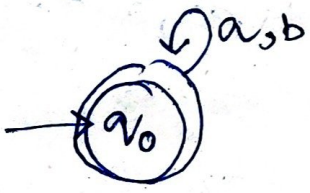
δ	a
→ q_0	q_1
* q_1	q_0

Q) Design a DFA to accept strings of a's & b's of any length

→ $\Sigma = \{a, b\}$

$L = \{\varepsilon, a, b, aa, aaa \ldots, bb, bbb \ldots, ab, ba \ldots\}$

min length → zero



$\Sigma = \{a, b\}$

$q_0 = q_0$ start

$F = q_0$

$Q = \{q_0\}$

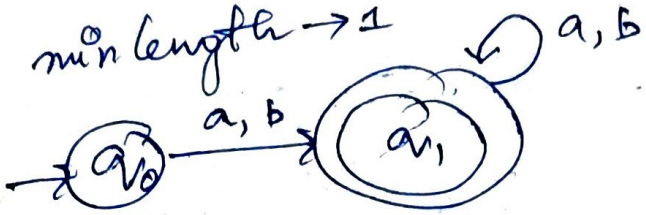
δ	a	b
* → q_0	q_0	q_0

a) at least 1 a's or 1 b's

$\Sigma = \{a, b\}$ $L = \{a, b, aa, aaa, ab, ab \ldots\}$

→

min length → 1



q_0 - start

$Q = \{q_0, q_1\}$

$F = q_1$

$\Sigma = \{a, b\}$

δ	a	b
→q_0	q_1	q_1
*q_1	q_1	q_1

a) Construct strings of a's & b's of even length

$L = \{\varepsilon, aa, bb, abb \ldots\}$

→

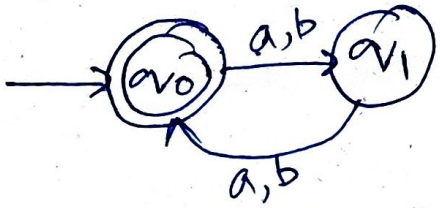
$\Sigma = \{a, b\}$

min string = zero

q_0 - start

$F = q_0$

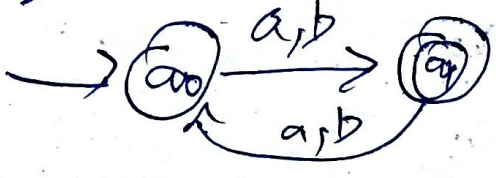
$Q = \{q_0, q_1\}$



δ

δ	a	b
→*q_0	q_1	q_1
q_1	q_0	q_0

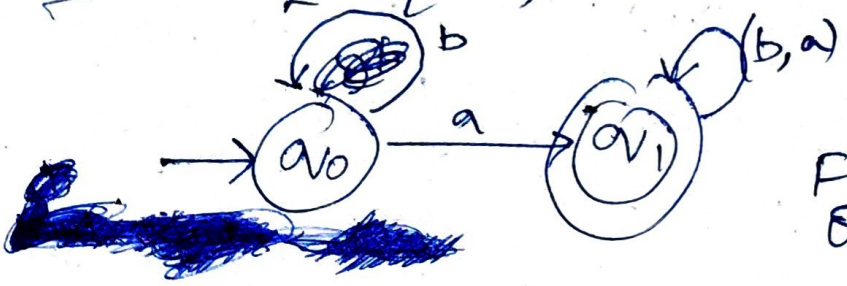
a) strings of as, bs of odd length



Q] Draw a DFA to accept strings of a's & b's having at least 1 a

Ans $\Sigma = \{a, b\}$

$L = \{a, aab, ab \ldots\}$



$q_0 \rightarrow$ initial

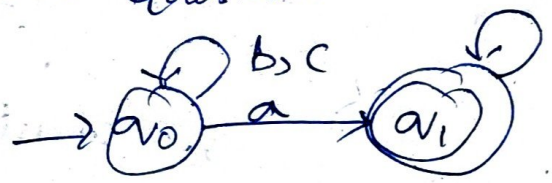
$F = q_1 \rightarrow$ Final

$Q = \{q_0, q_1\}$

δ	a	b
$\rightarrow q_0$	q_1	q_0
$*q_1$	q_1	q_1

Q] previous Question a's b's c's & at least 1a

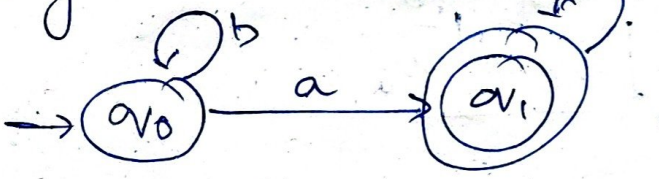
Ans



δ	a	b	c
$\rightarrow q_0$	q_1	q_0	q_0
$*q_1$	q_1	q_1	q_1

Q] DFA to accept strings of a's & b's exactly 1a

Ans



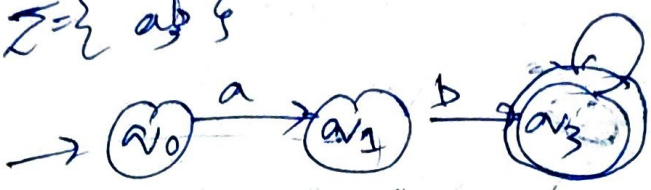
$L = \{a, ab, ba, bba\}$

δ	a	b
$\rightarrow q_0$	q_1	q_0
$*q_1$	ϕ	q_1

Q] DFA obtain strings of a's & b's starting with min length-2 & 3 states

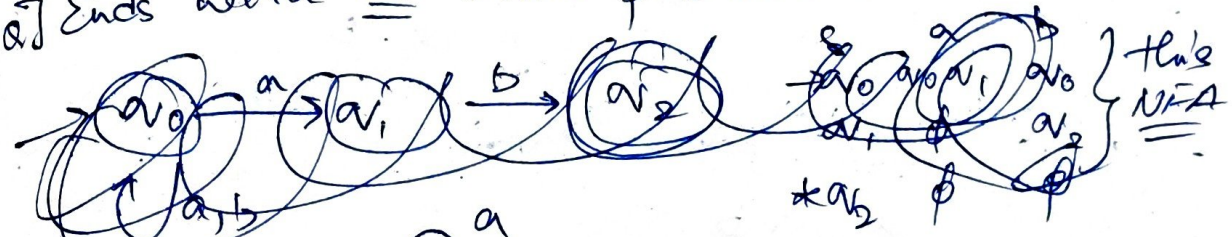
ab

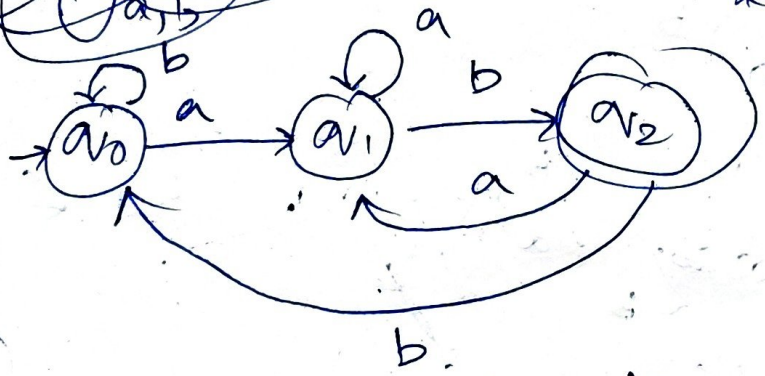
$\Sigma = \{a, b\}$



δ	a	b
$\to q_0$	q_1	ϕ
q_1	ϕ	q_3
$*q_3$	q_3	q_3

Q] Ends with ab same previous question.

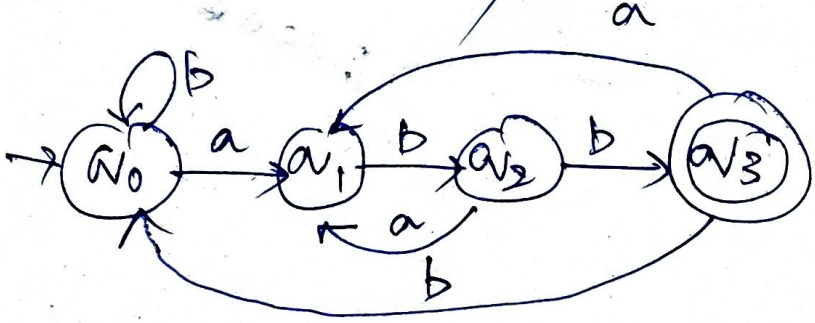




δ	a	b
$\to q_0$	q_1	q_0
q_1	q_1	q_2
$*q_2$	q_1	q_0

Q] Same Question abb ends

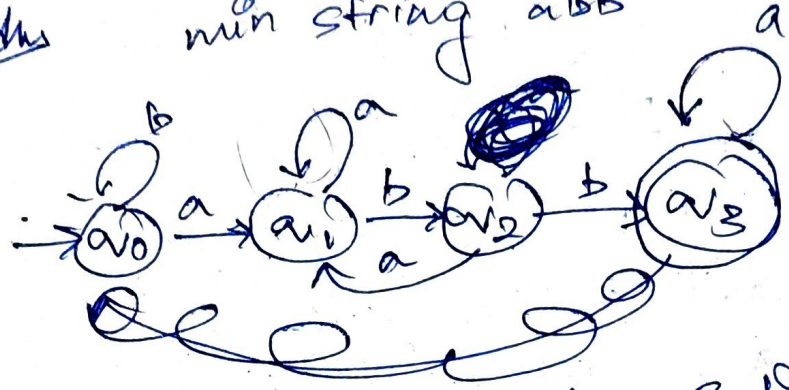
δ	a	b
$\to q_0$		
q_1		
q_2		
$*q_3$		



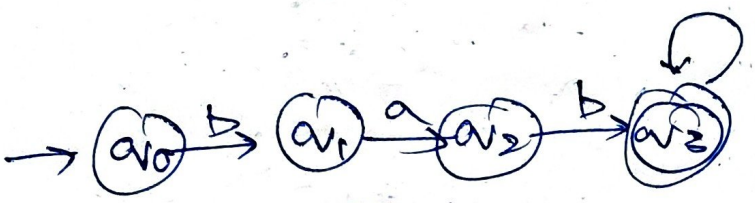
a)

a) Construct DFA with substring abb

Ans min string abb

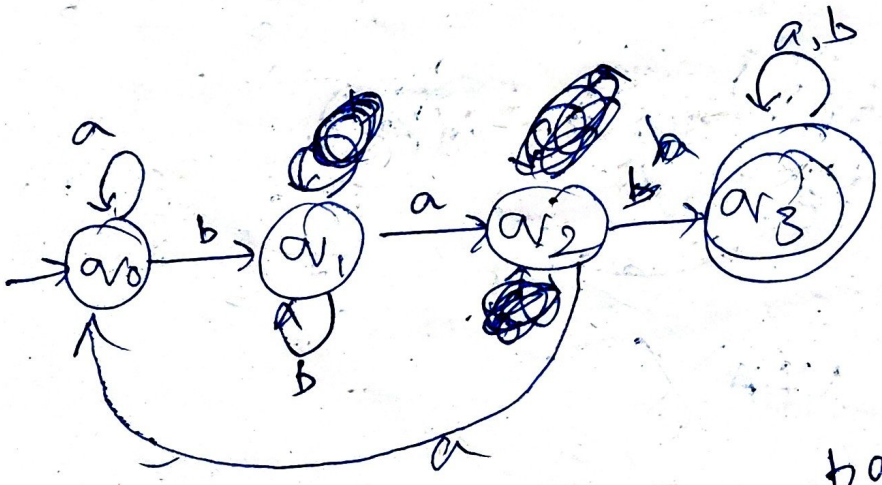


a) beginning with bab ending bab & substring



Begin.

End

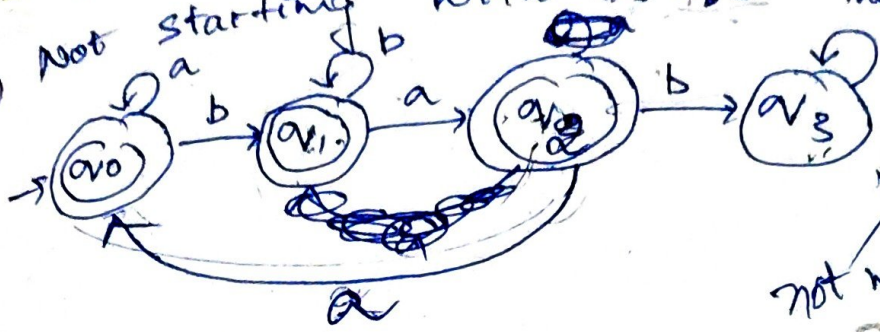


Substring

baaab

with Σ = a, b min string = 3

Q) Not starting



not working

baab

baaab

abab

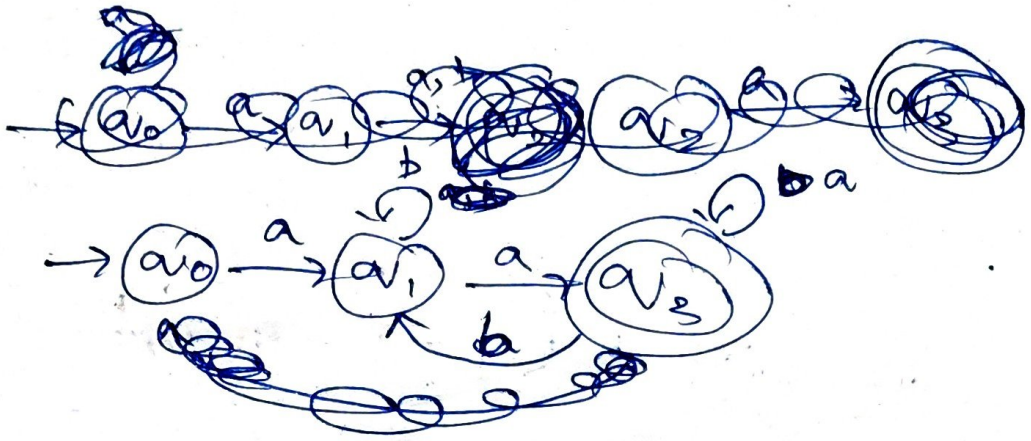
a

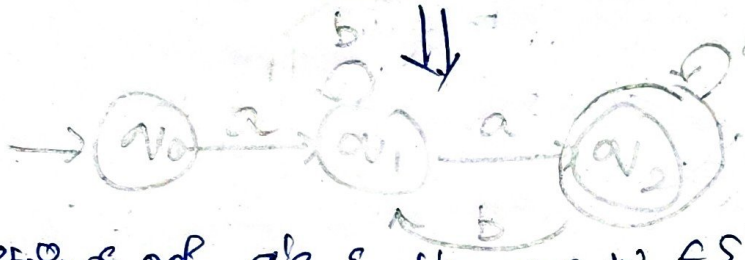
abbbb

Q) draw a DFA to accept strings of a's & bb such that the language $L = \{awa \mid w \in (a+b)^*\}$ where $n \geq 0$

awa $w \in (a+b)^* n$

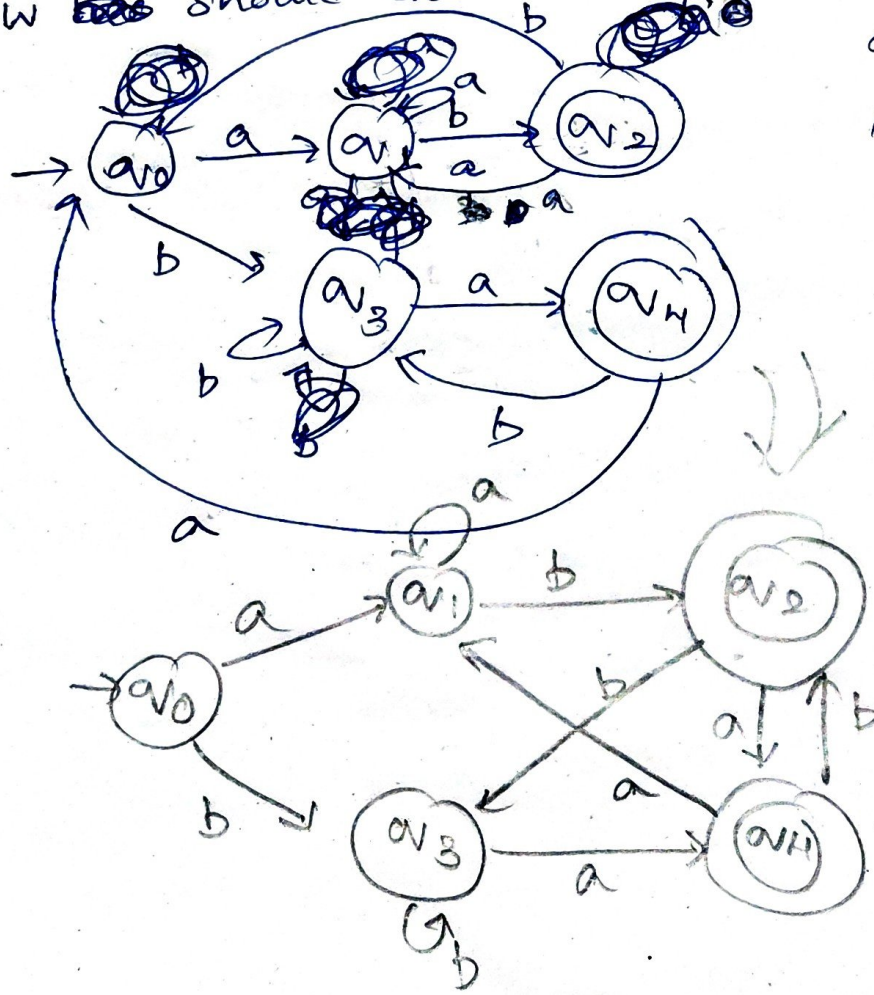
+ → or
. → and



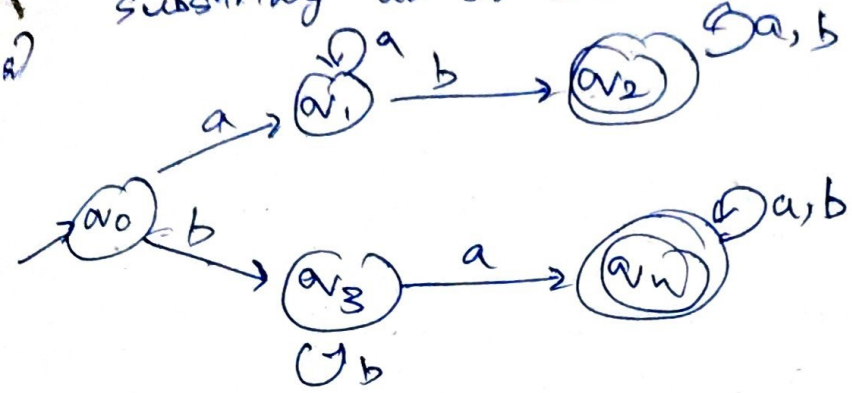


a) strings of a's & bb or $w \in \{a, b\}$ such that w should end with ab or ba or both

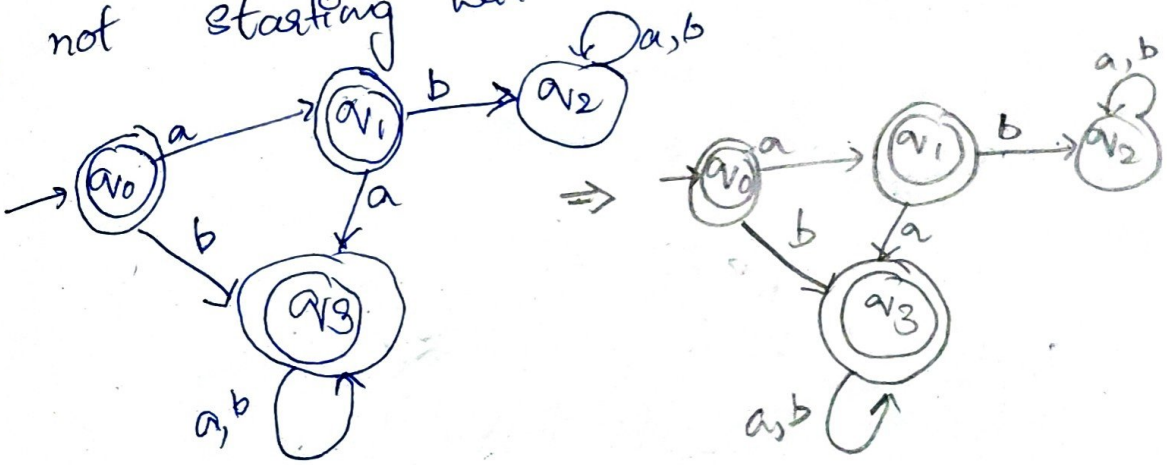
abab bu abab
bbbbb ab



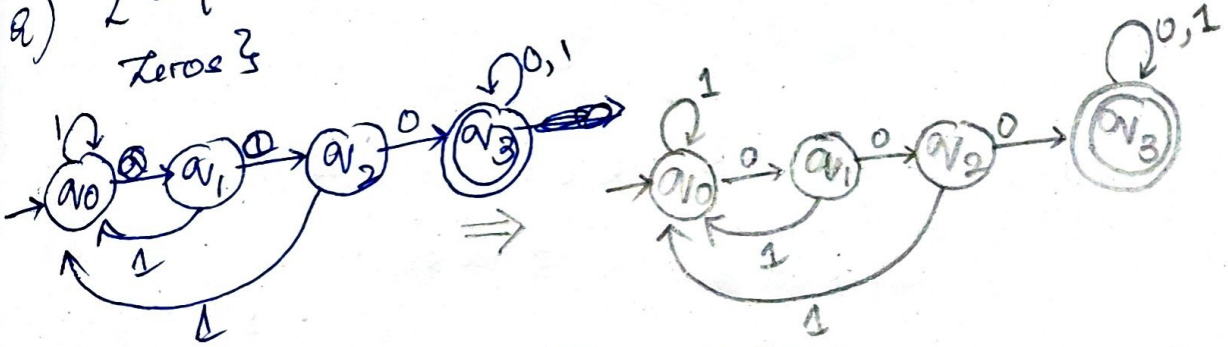
substring ab or ba



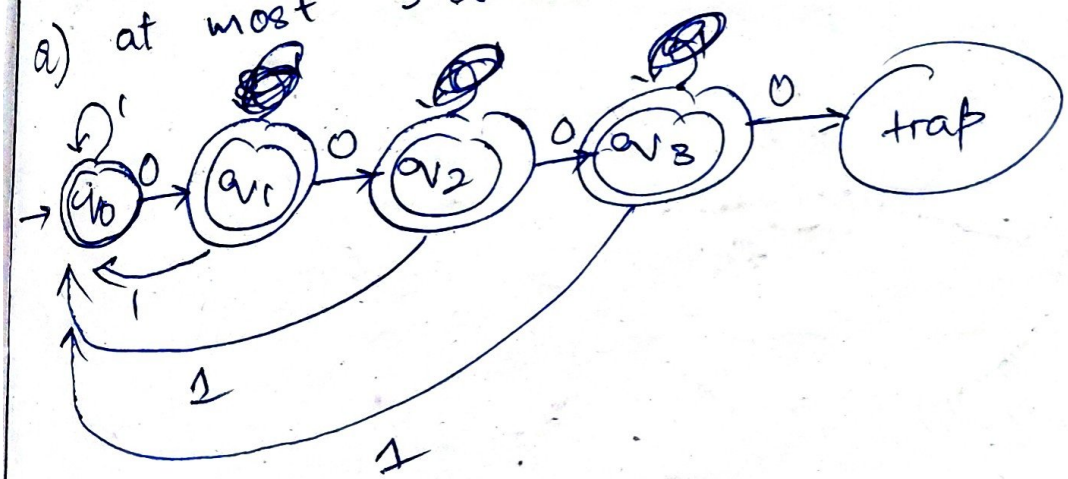
q) not starting with ab



q) $L = \{ w \in \{0,1\}^* \mid w$ with 3 consecutive zeros $\}$



q) at most 3 consecutive zeros

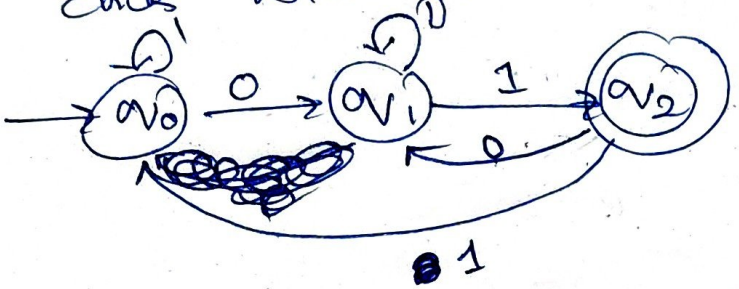


Q) $\Sigma \in (0,1)$ either begins or ends or both with 01

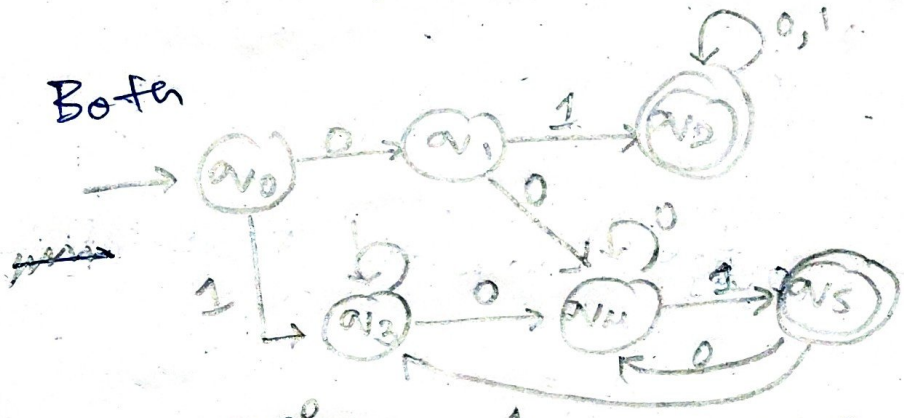
→ begin with 01

01 01

Ends with 01



Both

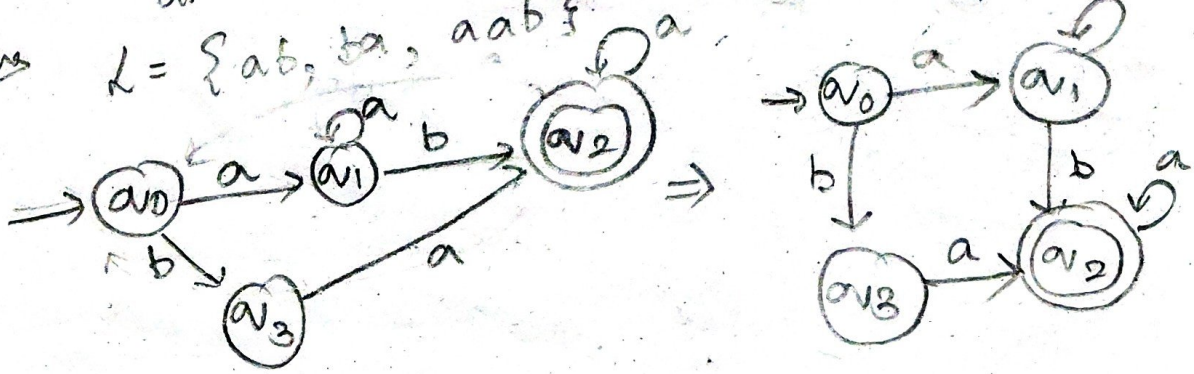


Q) $L = \{w \in$ number of a's $n_a(w) \leq 3 \quad w \in \{a,b\}^*\}$

→ at most 3 a's

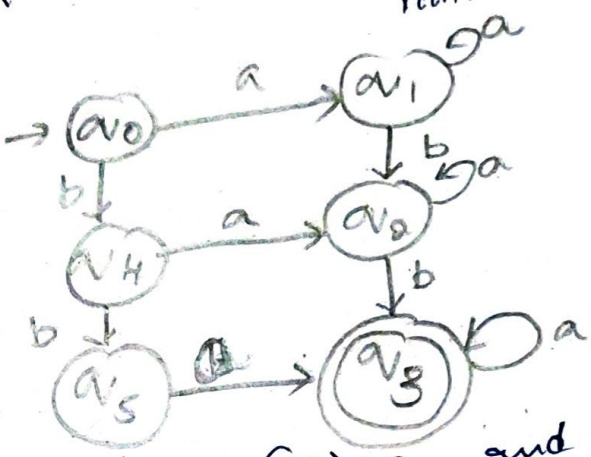
Q) $L = \{w \in n_a(w) \geq 1$ and $n_b(w) = 1 \quad w \in \{a,b\}^*\}$
at least 1 a with one b exact

Sol $L = \{ab, ba, aab\}$

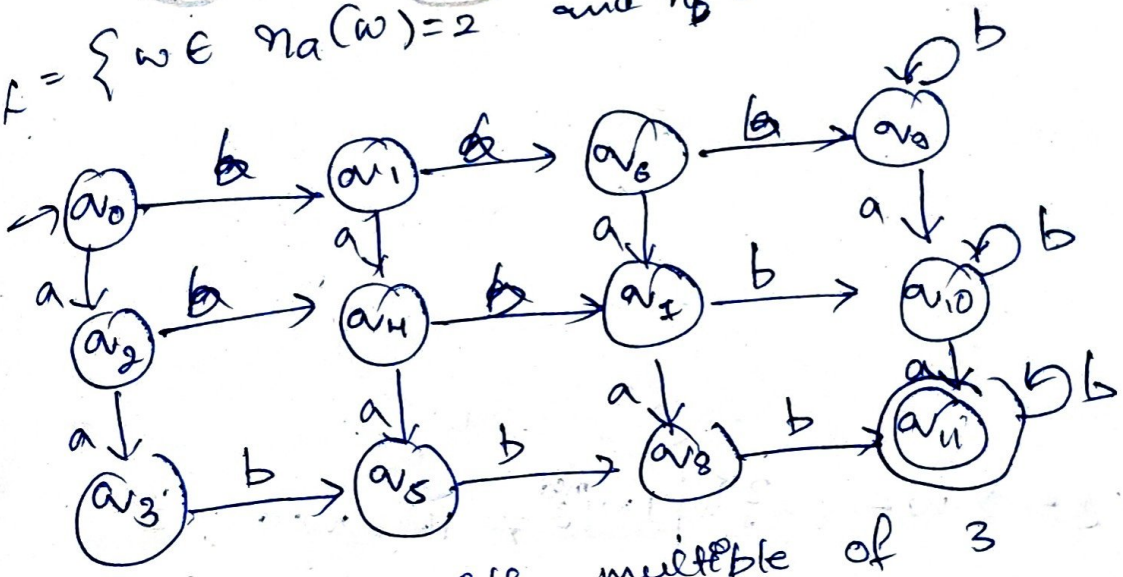


a) $L = \{w \in n_a(w) \geq 1$ and $n_b(w) = 2$ $w \in \{a,b\}^*\}$

min string: abb, bab, bba

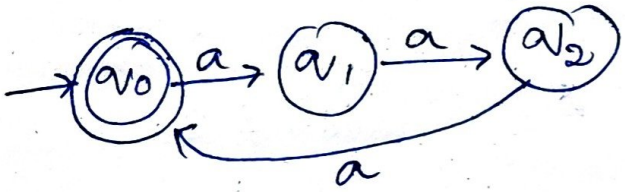


a) $L = \{w \in n_a(w) = 2$ and $n_b(w) \geq 3$ $w \in \{a,b\}^*\}$

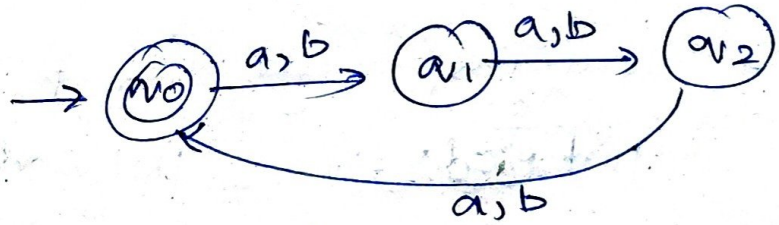


2) Strings of a's with multiple of 3

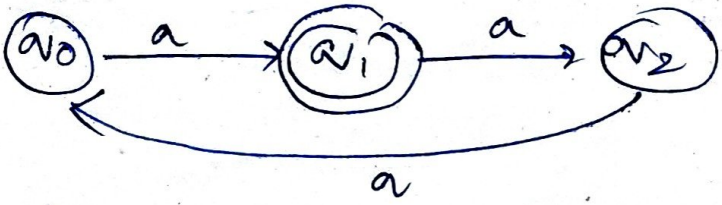
$w \in \{a\}^* \mid |w| \bmod 3 = 0$



a) Strings of a's & b's multiple of 3

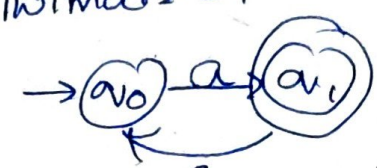


a) $w \in \{a\}^* \mid |w| \bmod 3 = 1$

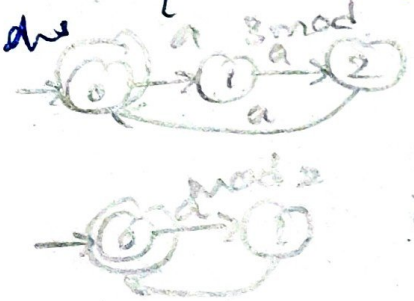


Q) $w \in \{a\}^* \mid |w| \bmod 2 = 0$

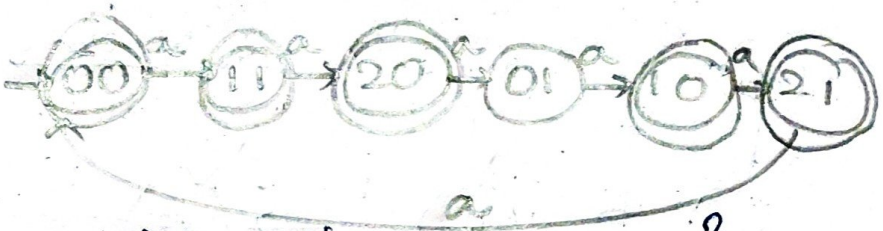
$|w| \bmod 2 = 1$



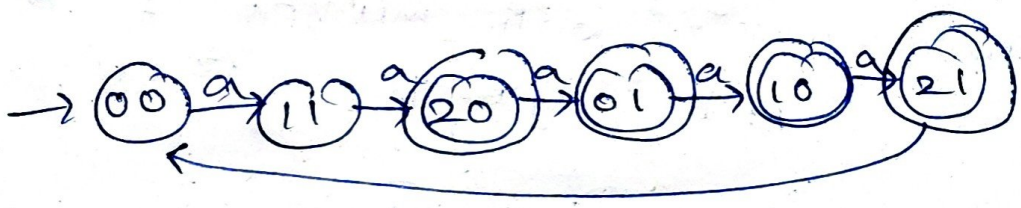
Q) $L = \{ w \in \{a\}^* \mid |w| \bmod 3 \geq |w| \bmod 2 \}$



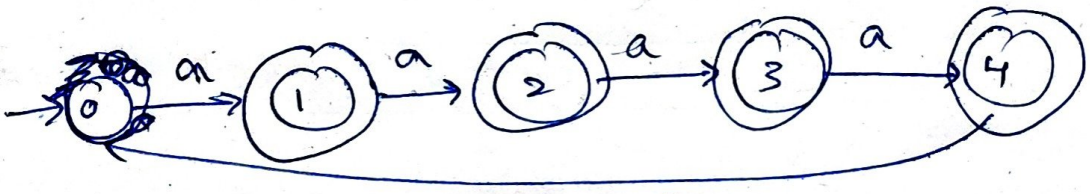
Cartesian product



Q) $L = \{ w \in \{a\}^* \mid |w| \bmod 3 \neq |w| \bmod 2 \}$

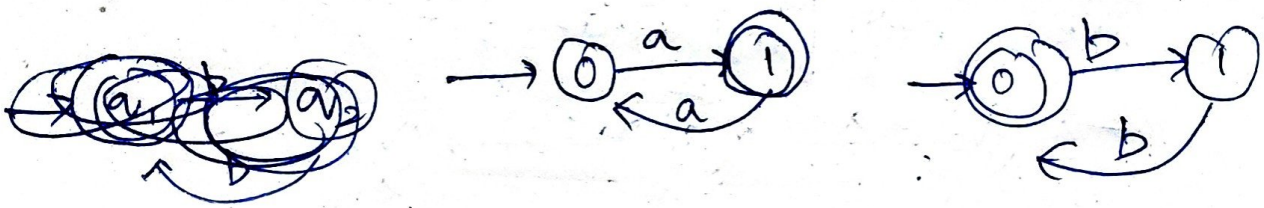


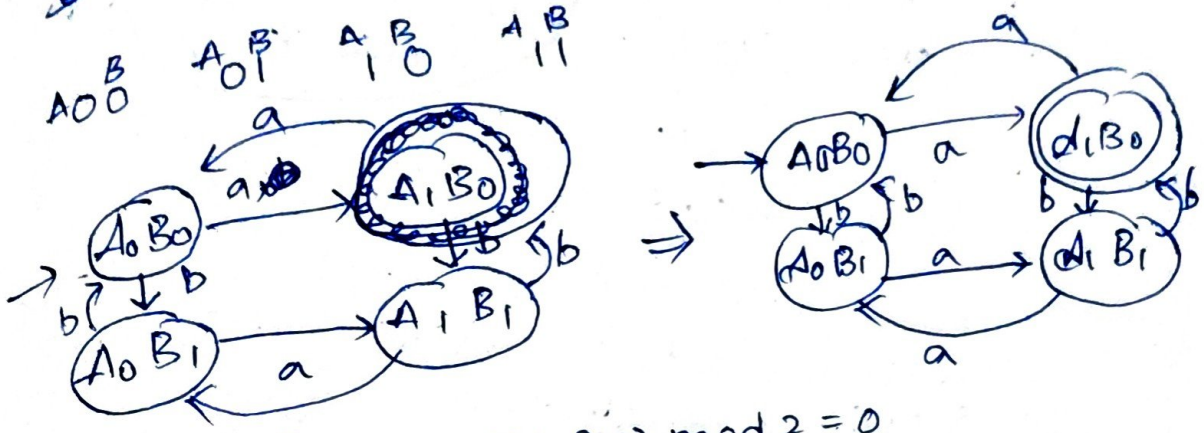
Q) obtain $w \in \{a\}^*$ $|w| \bmod 5 \neq 0$



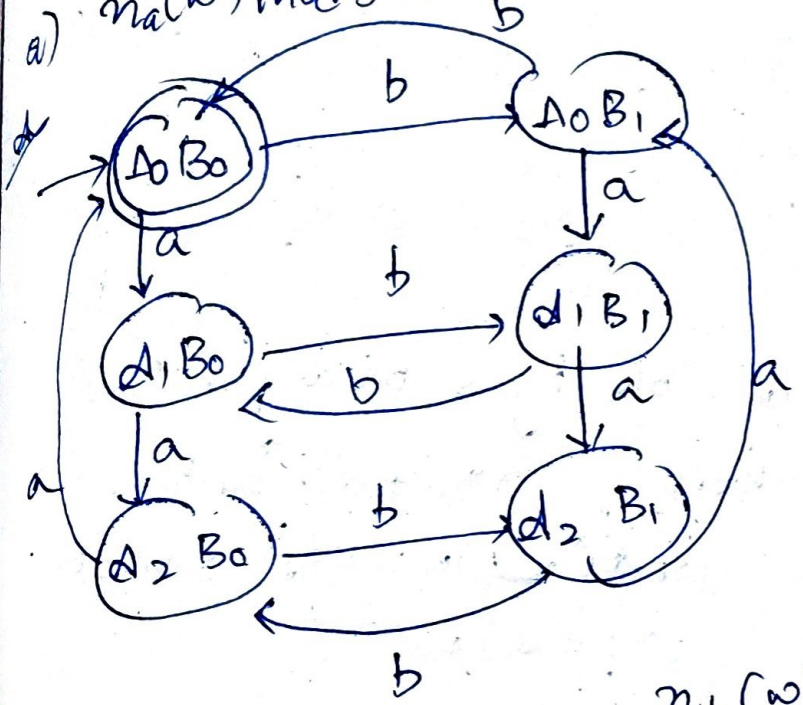
Q) obtain DFA contains odd a's & even b's

$|w| \bmod 2 = 1$ $\quad$ $|w| \bmod 2 = 0$

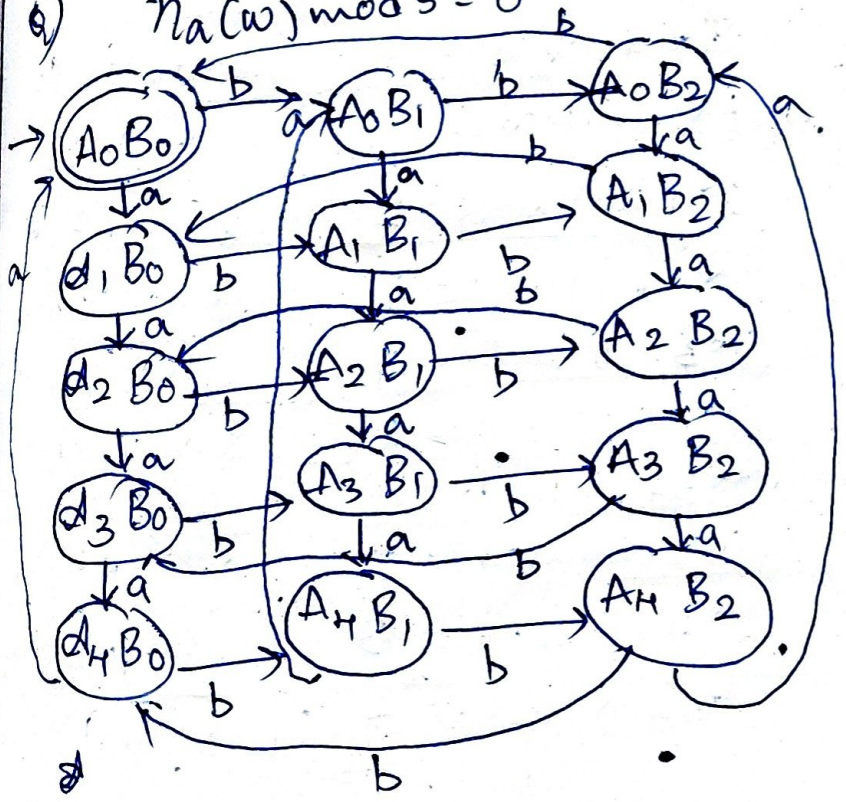




a) $n_a(w) \bmod 3 = 0$ $\quad n_b(w) \bmod 2 = 0$



b) $n_a(w) \bmod 5 = 0$ $\quad n_b(w) \bmod 3 = 0$



Q) Construct a DFA considers strings of 0's & 1's the values of each string is represented as binary numbers (0) mod 5 should be accepted

→

↓

mod 5 = 0

[radix $r = 2 \rightarrow$ binary
$r = 8 \rightarrow$ octal
$r = 16 \rightarrow$ hexadecimal
$r = 10 \rightarrow$ decimal]

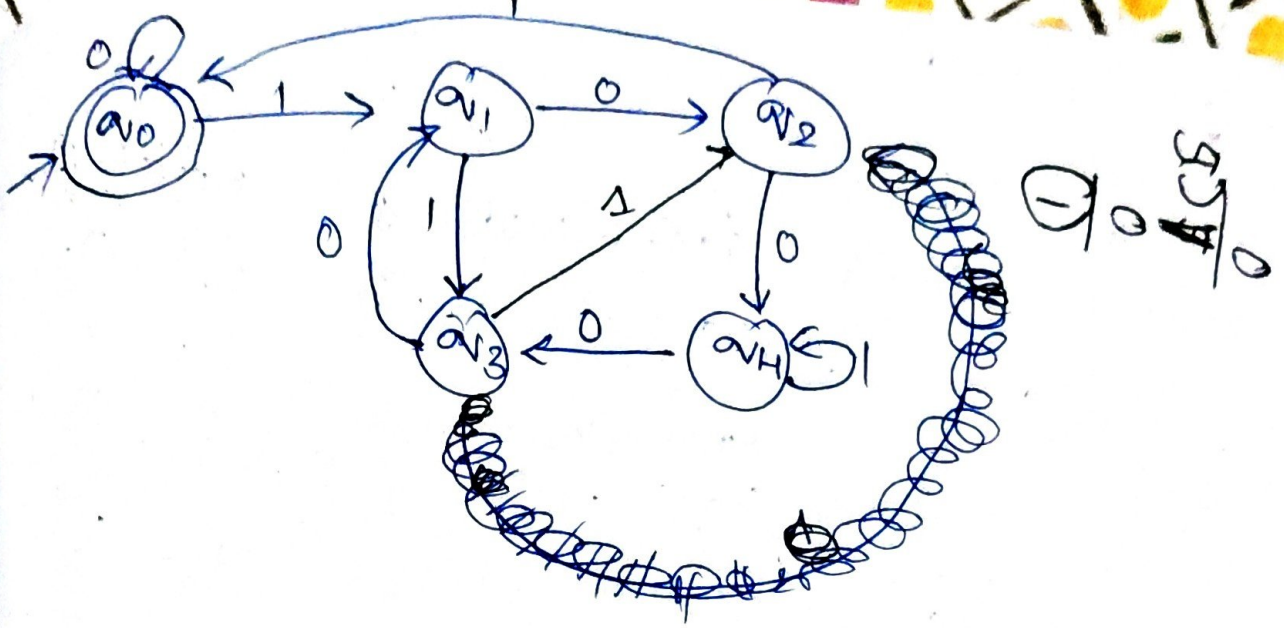
[digit $d = \{0, 1\} \rightarrow$ binary
$d = \{0, 1, 2 \ldots 7\} \rightarrow$ octal
$d = \{0, 1, 2 \ldots 9\} \rightarrow$ decimal]

$k = 5$ } → divisible by 5 binary values

Steps

$\delta(a_i, d) = a_j$ where $j = (r * i + d) \bmod k$

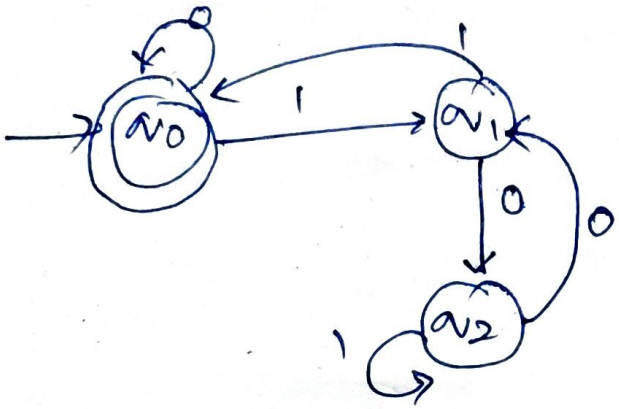
reminder i {0,1,2,3,4}	input {0,1}	$(r * i + d) \bmod k = j$ $(2 * i + d) \bmod 5 = j$	$\delta(a_i, d) = a_j$
$i = 0$	0	$(2*0+0) \bmod 5 = 0$	$\delta(a_0, 0) = 0\ a_0$
	1	$(2*0+1) \bmod 5 = 1$	$\delta(a_0, 1) = 1\ a_1$
$i = 1$	0	$(2*1+0) \bmod 5 = 2$	$\delta(a_1, 0) = 2\ a_2$
	1	$(2*1+1) \bmod 5 = 3$	$\delta(a_1, 1) = 3\ a_3$
$i = 2$	0	$(2*3+0) \bmod 5 = 4$	$\delta(a_2, 0) = 4\ a_4$
	1	$(2*2+1) \bmod 5 = 0$	$\delta(a_2, 1) = 0\ a_0$
$i = 3$	0	$(2*3+0) \bmod 5 = 1$	$\delta(a_3, 0) = 1\ a_1$
	1	$(2*3+1) \bmod 5 = 2$	$\delta(a_3, 1) = 2\ a_2$
$i = 4$	0	$(2*4+0) \bmod 5 = 3$	$\delta(a_4, 0) = 3\ a_3$
	1	$(2*4+1) \bmod 5 = 4$	$\delta(a_4, 1) = 4\ a_4$



Bin Number	Decimal	$k = 5$
0 0 0	0	q_0
0 0 1	1	q_1
0 1 0	2	q_2
0 1 1	3	q_3
1 0 0	4	q_4
1 0 1	5	q_0
1 1 0	6	q_1
1 1 1	7	q_2
1 0 0 0	8	q_3
1 0 0 1	9	q_4
1 0 1 0	10	q_0

do mod 3 = 0

reminder {i}	input {0, 1}	(2*i+d) mod 3 = j	$\delta(a_i, d) = a_j$
0	0	(2*0+0) mod 3 = 0	$a_0 \xrightarrow{0} a_0$
	1	(2*0+1) mod 3 = 1	$a_0 \xrightarrow{1} a_1$
1	0	(2*1+0) mod 3 = 2	$a_1 \xrightarrow{0} a_2$
	1	(2*1+1) mod 3 = 0	$a_1 \xrightarrow{1} a_0$
2	0	(2*2+0) mod 3 = 1	$a_2 \xrightarrow{0} a_1$
	1	(2*2+1) mod 3 = 2	$a_2 \xrightarrow{1} a_2$



decimal divisible by 3

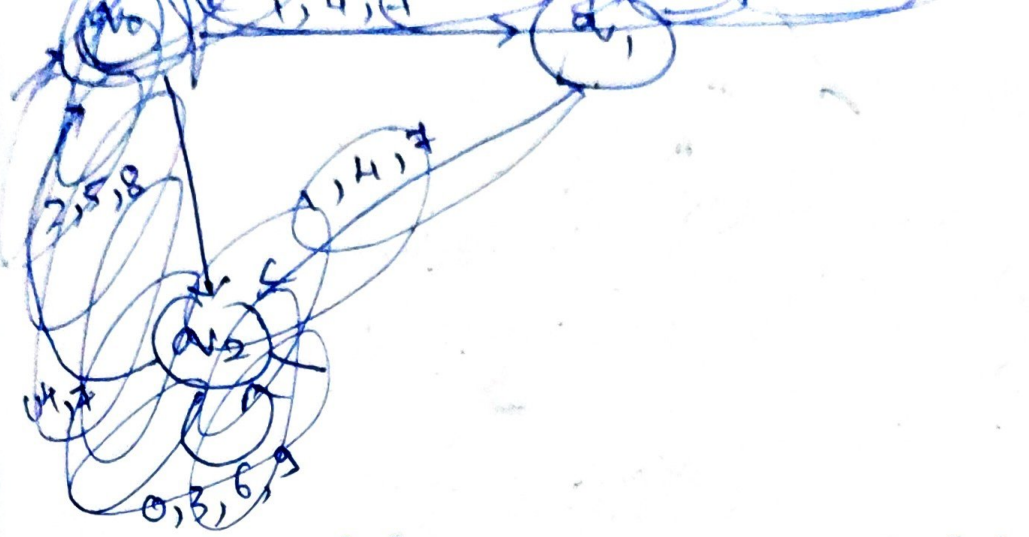
$a_0 \rightarrow$ 0 3 6 9

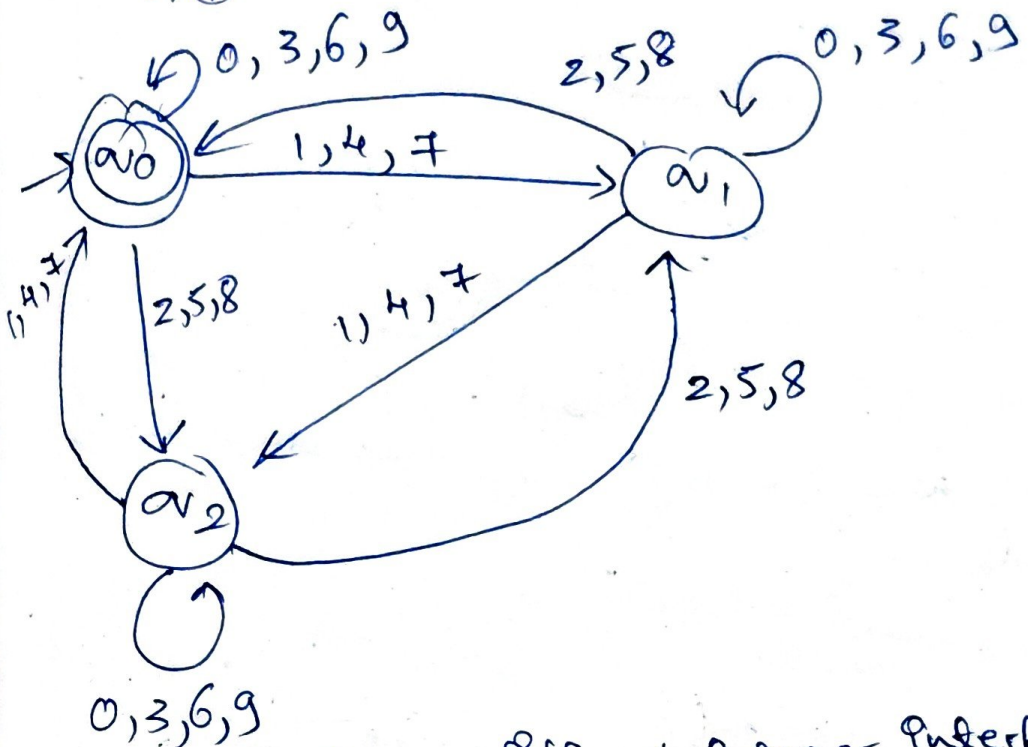
$a_1 \rightarrow$ 1 4 7

$a_2 \rightarrow$ 2 5 8

(10*i+d) mod 3

reminder {i}	input		
{0,1,2}	0 3 6 9		
i=0	1 4 7		
	2 5 8		
	0 3 6 9	(10x0+0) mod 3 = 0	$a_0 \xrightarrow{0,3,6,9} a_0$
	1 4 7	(10x0+1) mod 3 = 1	$a_0 \xrightarrow{1,4,7} a_1$
	2 5 8	(10x0+2) mod 3 = 2	$a_0 \xrightarrow{2,5,8} a_2$
i=1	0 3 6 9	(10x1+0) mod 3 = 1	$a_1 \rightarrow a_1$
	1 4 7	(10x1+1) mod 3 = 2	$a_1 \rightarrow a_2$
	2 5 8	(10x1+2) mod 3 = 0	$a_1 \rightarrow a_0$
i=2	0 3 6 9	(10x2+0) mod 3 = 2	$a_2 \rightarrow a_2$
	1 4 7	(10x2+1) mod 3 = 0	$a_2 \rightarrow a_0$
	2 5 8	(10x2+2) mod 3 = 1	$a_2 \rightarrow a_1$

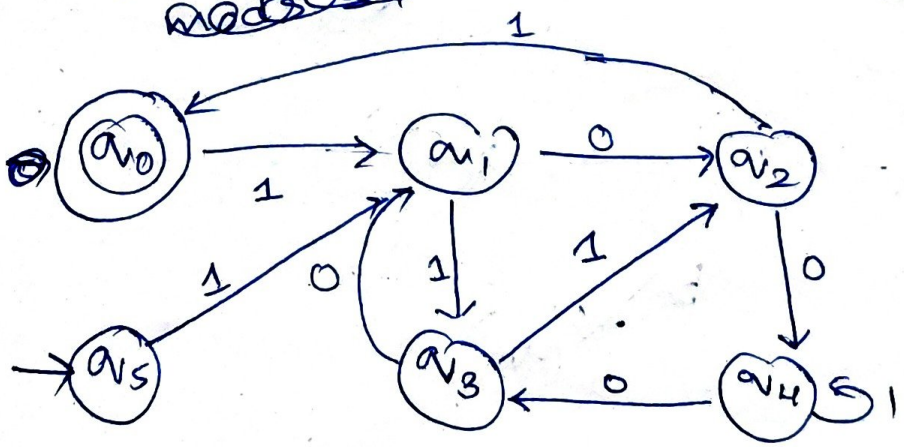




Q) DFA Begining with 1 integer interpreted as binary & multiplied of 5

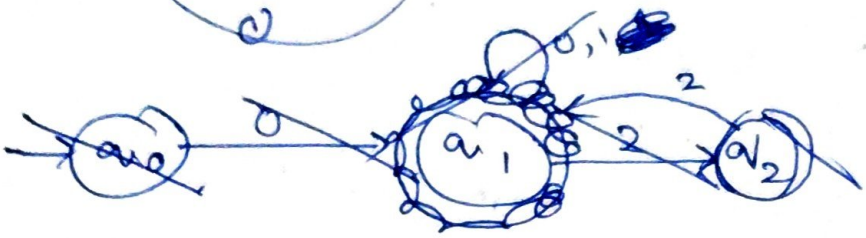
mod 5 = 0 not start with 1

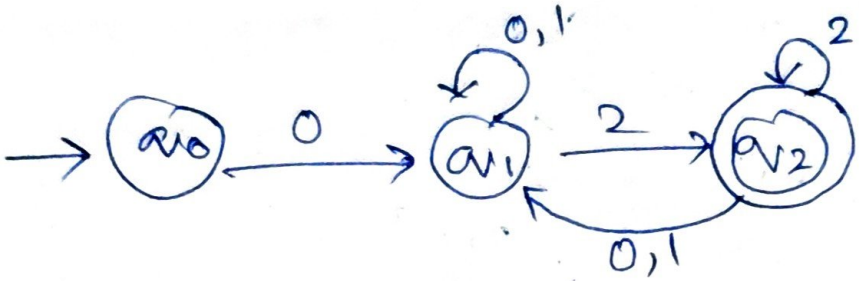
Ans



Q) DFA $w = 0s2$ $s \in \{0,1,2\}^*$

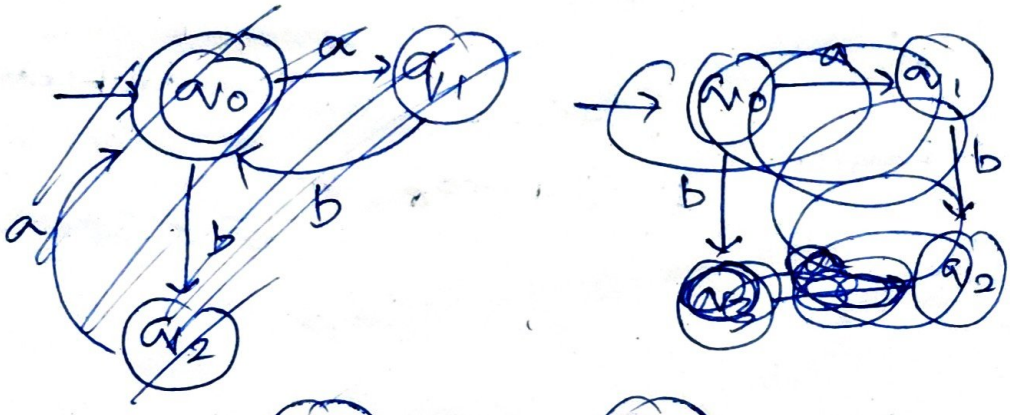
Sol $\varepsilon \rightarrow$ accepted

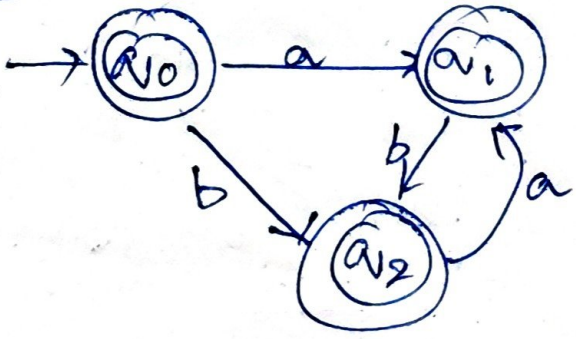




Q) DFA $w \in \{a,b\}^*$ no two consecutive Symbols

Sol





NFA → Non-deterministic finite automaton

NFA is a Quine tuple or 5 tuple

$M = (Q, \Sigma, \delta, q_0, F)$

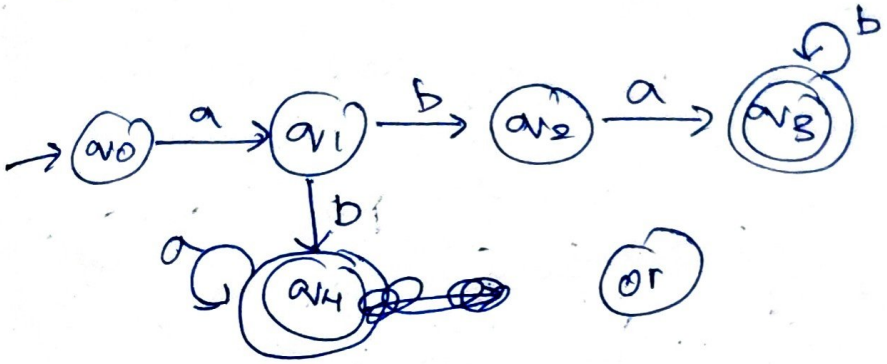
$Q \rightarrow$ Finite set of states

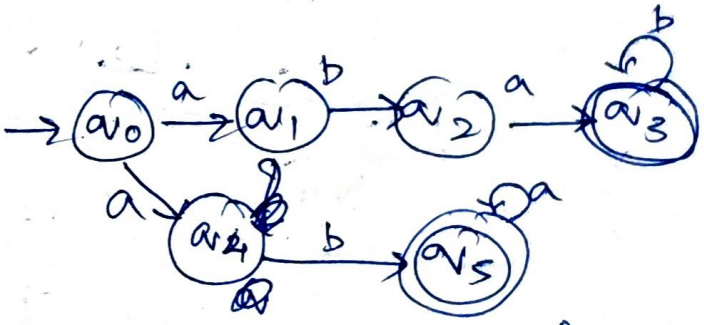
$\Sigma \rightarrow$ alphabet

$F \subseteq Q$

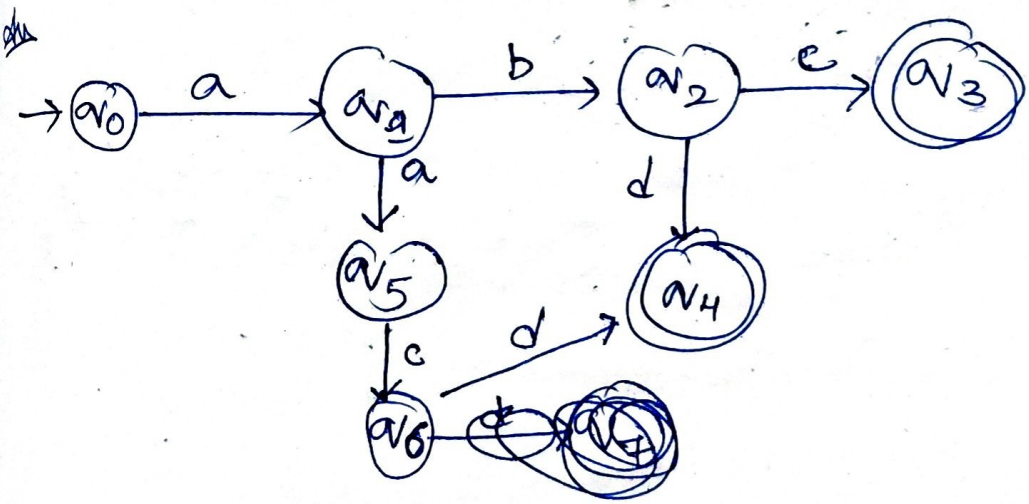
$\delta \rightarrow Q \times \Sigma$ to 2^Q

a) $L = \{w \mid w \in \{aba b^n \text{ or } ab a^n \text{ where } n >= 0\}$



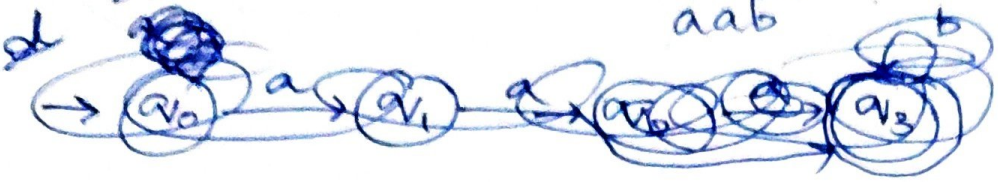


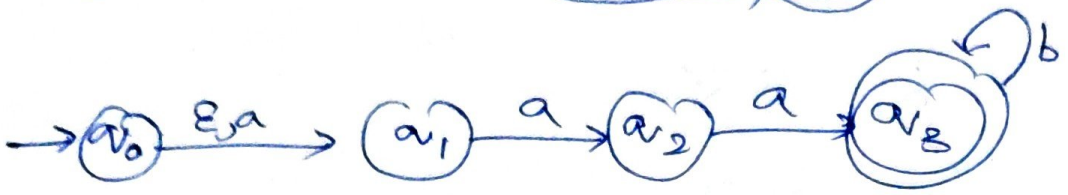
b) NFA to Recognize the following set of strings. abc, abd & aacd



Q) $w \in \{a,b\}^* /$ w is made up of an optional a followed by a a followed by 0 or more b's

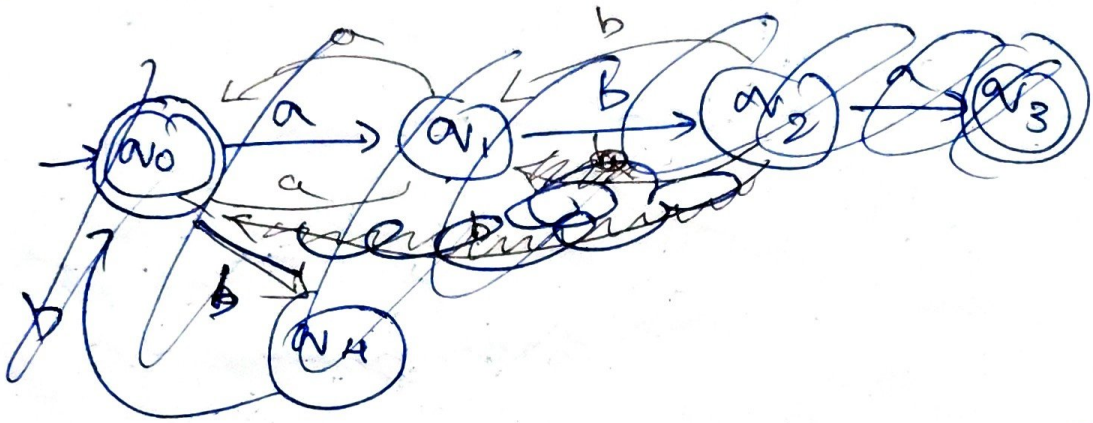
Sol

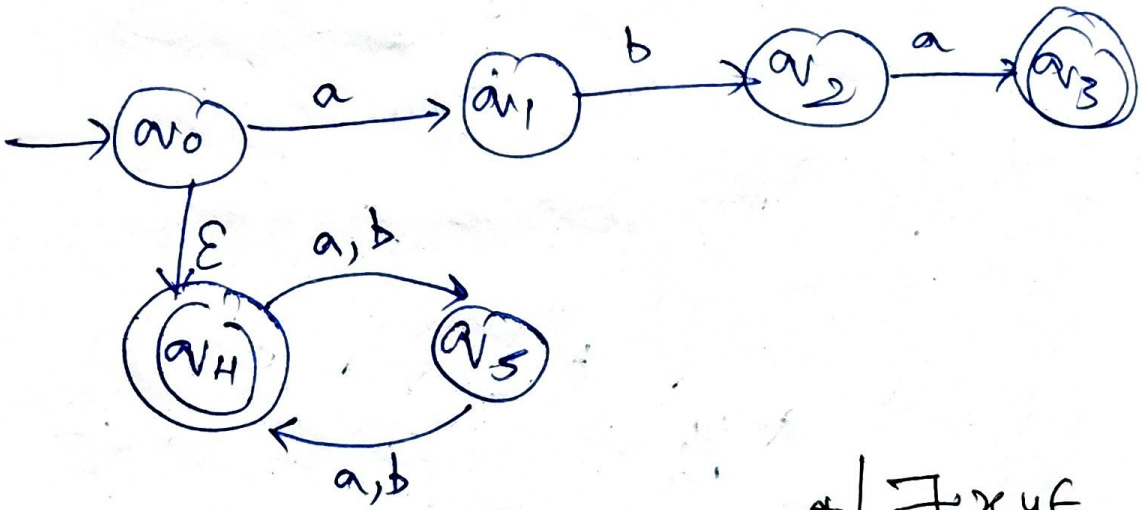




Q) $w \in \{a,b\}^* /$ w = aba or |w| is even

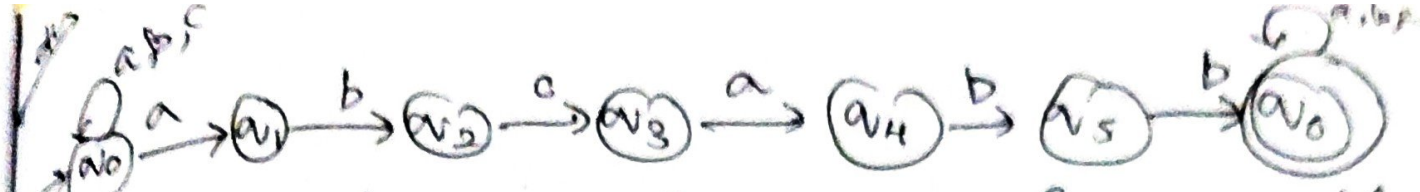
Sol





Q) Design a NFA $L = \{w \in \{a,b,c\}^* / \nexists x,y \in \{a,b,c\}^*$

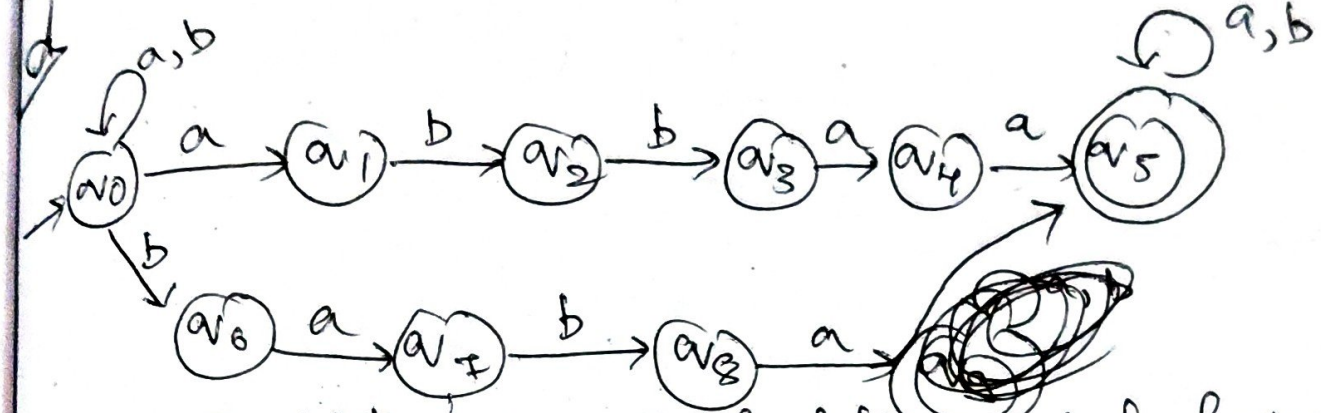
$w = xabcabby\}$



a) $w \in \{a,b\}^* \mid \exists x,y \in \{a,b\}^* \quad w = (x\,abb\,aa\,y)$

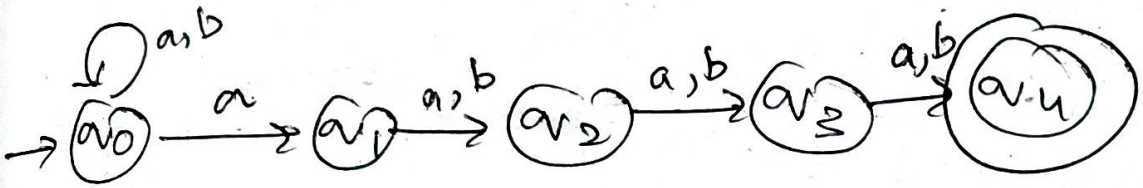
$(w = xbabay)$

or



a) $w \in \{a,b\}^*$ ~~[illegible]~~ 'fourth symbol from last character is a

$(a+b)^*\ a\ (a+b)\ (a+b)\ (a+b)$

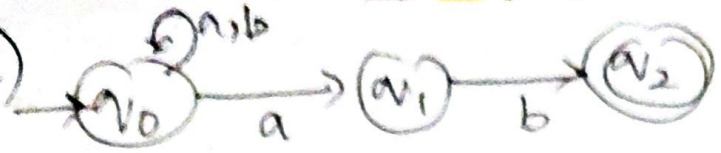


CONVERSION FROM NFA to DFA

① Subset construction Method

② Lazy Evaluation Method

Q)



δ_N	a	b
→ q_0	$\{q_0, q_1\}$	q_0
q_1	ϕ	q_2
← q_2	ϕ	ϕ

} NFA

Step 1: Start state of DFA is start state of NFA (q_0)

Step 2: Identify Σ ($\Sigma = \{a, b\}$)

Step 3: Identify states of DFA

$Q_D = \{\phi, \{q_0\}, \{q_1\}, \{q_2\}, \{q_0, q_1\}, \{q_0, q_2\}, \{q_1, q_2\}, \{q_0, q_1, q_2\}\}$

Step 4: Identify the transitions

first state ϕ

$\delta_D(\phi, a) = \phi$ $\quad \delta_D(\phi, b) = \phi$

$\delta_D(\{q_0\}, a) = \{q_0, q_1\}$ $\quad \delta_D(\{q_0\}, b) = q_0$

$\delta_D(\{q_1\}, a) = \phi$ $\quad \delta_D(\{q_1\}, b) = q_2$

$\delta_D(\{q_2\}, a) = \phi$ $\quad \delta_D(\{q_2\}, b) = \phi$

$\delta_D(\{q_0, q_1\}, a) = \delta_D(\{q_0\}, a) \cup \delta_D(\{q_0, q_1\}, a)$

$= \{q_0, q_1\} \cup \phi = \{q_0, q_1\}$

$\delta_D(\{q_0, q_1\}, b) = \{q_0\} \cup \{q_2\} = \{q_0, q_2\}$

$\delta_D(\{q_0, q_2\}, a) = \{q_0\} \cup \{q_1\} = \{q_0, q_1\}$

$\delta_D(\{q_0, q_2\}, b) = \{q_0\} \cup \phi = \{q_0\}$

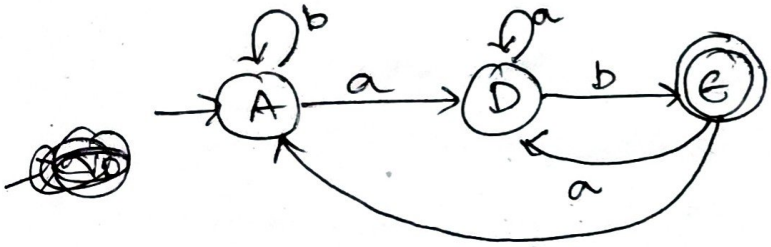
$\delta_D(\{q_1, q_2\}, a) = \phi \cup \phi = \phi$

$\delta_D(\{q_1, q_2\}, b) = \{q_2\} \cup \phi = \{q_2\}$

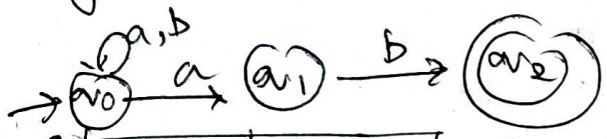
$\delta_D(\{q_0, q_1, q_2\}, a) = \{q_0, q_1\} \cup \phi \cup \phi = \{q_0, q_1\}$

$\delta_D(\{q_0, q_1, q_2\}, b) = \{q_0\} \cup \{q_2\} \cup \phi = \{q_0, q_2\}$

δ_D	a	b
ϕ	ϕ	ϕ
$\{q_0\}$ A	$\{q_0, q_1\}$ D	$\{q_0\}$ A
$\{q_1\}$ B	ϕ	q_2 C
$\{q_2\}$ C	ϕ	ϕ
$\{q_0, q_1\}$ D	$\{q_0, q_1\}$ D	$\{q_0, q_2\}$ E
$\{q_0, q_2\}$ E	$\{q_0, q_1\}$ D	$\{q_0\}$ A
* $\{q_1, q_2\}$ F	ϕ	$\{q_2\}$ C
$\{q_1, q_2, q_0\}$ G	$\{q_0, q_1\}$ D	$\{q_0, q_2\}$ E



Lazy Evaluation For Same Problem



δ	a	b
$\to q_0$	$\{q_0, q_1\}$	q_0
q_1	ϕ	q_2
* q_2	ϕ	ϕ

Steps: Start state q_0 for NFA

Step 2: $\Sigma = \{a, b\}$

Step 3: Transitions

$\{q_0\} = A$

$A: \delta_D(\{q_0\}, a) = \{q_0, q_1\} - B$

$A: \delta_D(\{q_0\}, b) = \{q_0\} - A$

Consider state $= \{q_0, q_1\} - B$

$B: \delta_D(\{q_0, q_1\}, a) = \delta_N(\{q_0\}, a) \cup \delta_N(\{q_1\}, a) = \{q_0, q_1\} \cup \phi = \{q_0, q_1\}$ B'

$B: \delta_D(\{q_0, q_1\}, b) = \delta_N(\{q_0\}, b) \cup \delta_N(\{q_1\}, b)$

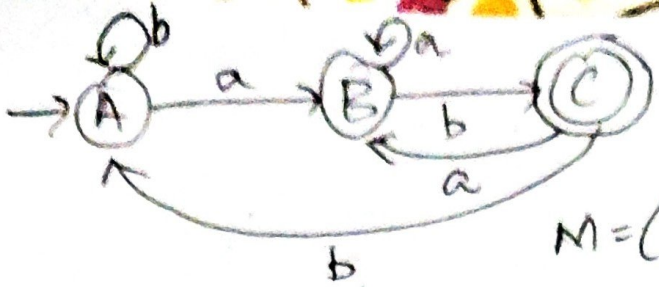
$= \{q_0\} \cup \{q_2\} = \{q_0, q_2\} - C$

Consider the new state $C \to \{q_0, q_2\}$

$C: \delta_D(\{q_0, q_2\}, a) = \delta_N(\{q_0\}, a) \cup \delta_N(\{q_2\}, a)$

$= \{q_0, q_1\} - B$

$C: \delta_D(\{q_0, q_2\}, b) = \{q_0\} - A$



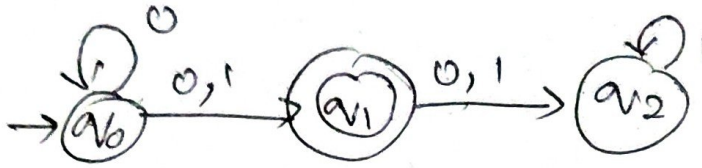
$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{A, B, C\}$

$q_0 = \text{start}$

$F = C$

$\Sigma = \{a, b\}$

a) 

NFS

δ	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	q_1
$* q_1$	q_2	q_2
q_2	ϕ	q_2

q_0 – start

$\Sigma = \{0, 1\}$

Transitions

$\{q_0\}$ — A

A : $\delta_D(\{q_0\}, 0) = (q_0, q_1) = B$

A : $\delta_D(\{q_0\}, 1) = q_1 - C$

B : $\delta_D(\{q_0, q_1\}, 0) = \{q_0, q_1\} \cup q_2$

$= \{q_0, q_1, q_2\} - D$

B : $\delta_D(\{q_0, q_1\}, 1) = \{q_1, q_2\} - E$

C : $\delta_D(\{q_1\}, 0\} = q_2 - F$

C : $\delta_D(\{q_1\}, 1\} = q_2 - F$

D : $\delta(\{q_0, q_1 q_2\}, 0) = \{q_0 q_1\} \cup q_2 \not\in \phi = D$

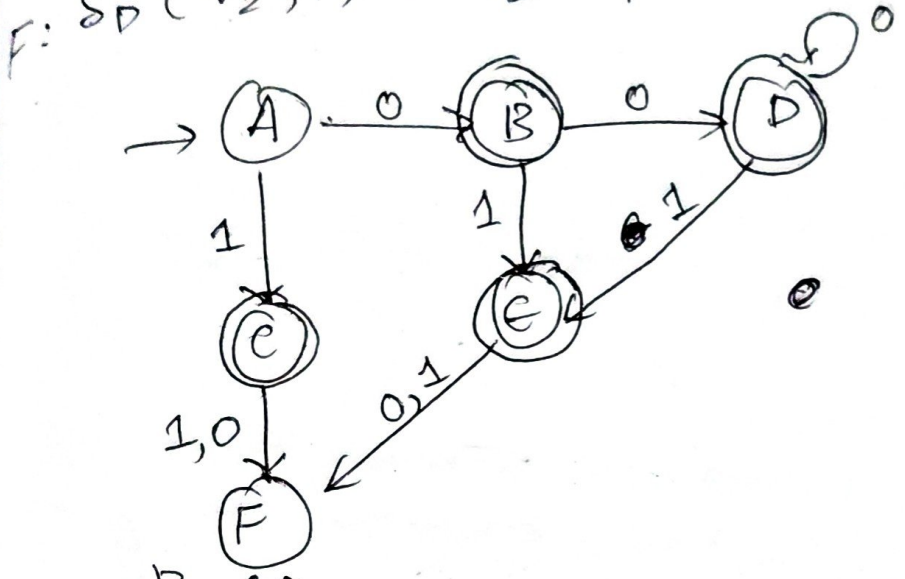
D : $\delta(\{q_0, q_1, q_2\}, 1) = \{q_1, q_2\} - E$

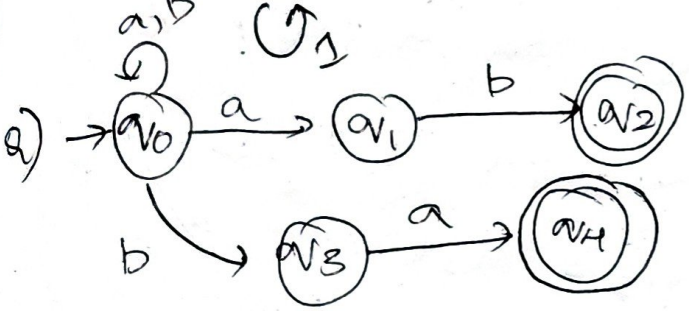
E: $\delta_D(q_1, q_2, 0) = \{q_2, \phi\}$ – F

E: $\delta_D((q_1, q_2)\ 1\} = q_2$ – F

F: $\delta_D(q_2, \phi) = \phi$

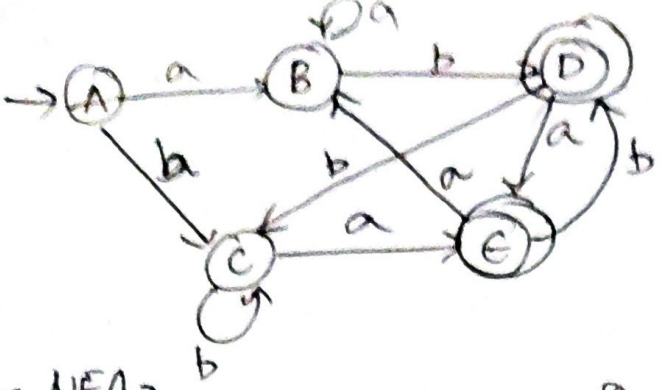
F: $\delta_D(q_2, 1) = q_2$ – F





δ	a	b
→q_0	$\{q_0, q_1\}$	$\{q_1, q_3\}$
q_1	ϕ	q_2
*q_2	ϕ	ϕ
q_3	q_4	ϕ
*q_H	ϕ	ϕ

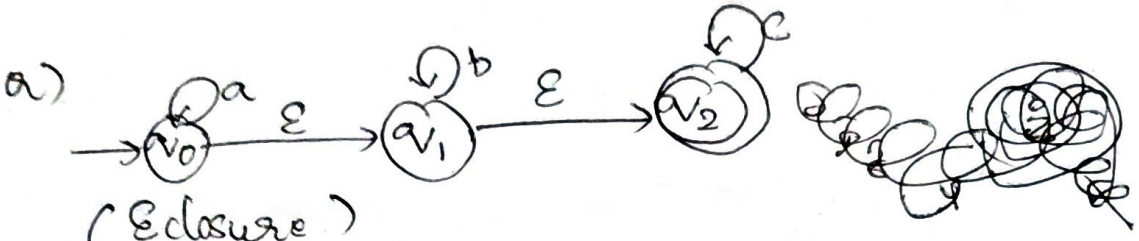
δ	a	b
	B	C
A: $\{q_0\}$	$\{q_0, q_1\}$	$\{q_0, q_3\}$
B: $\{q_0, q_1\}$	$\{q_0, q_1\}$: B	$\{q_0, q_2, q_3\}$: D
C: $\{q_0, q_3\}$	$\{q_0, q_1, q_H\}$: E ✓	$\{q_0, q_3\}$: C
D: $\{q_0, q_2, q_3\}$	$\{q_0, q_1, q_H\}$: E	$\{q_0, q_3, \}$: C
E: $\{q_0, q_1, q_H\}$	$\{q_0, q_1\}$: B	$\{q_0, q_3, q_2\}$: D



Epsilon NFA to DFA Conversion

ε-NFA to DFA

a)



(ε closure)
(ε - closure)

Step 1: To find start state ~~of~~ dfa ~~to~~ from ε-NFA we have to find ε-closure

ε-closure: $v_0 = \{v_0, v_1, v_2\} = A$

ε-closure: $v_1 = \{v_1, v_2\}$

ε-closer : $v_2 = \{v_2\}$

Consider state $A = \{v_0, v_1, v_2\}$ ↓ apply ε closure

ε-closure $(\{v_0, v_1, v_2\}, a) = \{v_0\} = \{v_0, v_1, v_2\}$
$= A$

ε-closure $(\{v_0, v_1, v_2\}, b) =$ ε-closure $\{v_1\}$
$= \{v_1, v_2\} - B$

ε-closure $(\{v_0, v_1, v_2\}, c) =$ ε-closure $\{v_2\}$
$= \{v_2\} - C$

B - $\{q_1, q_2\}$

ε-closure (B, a) = $\emptyset$

ε-closure (B, b) = $\{q_1\}$ - B = $\{q_1, q_2\}$ = B

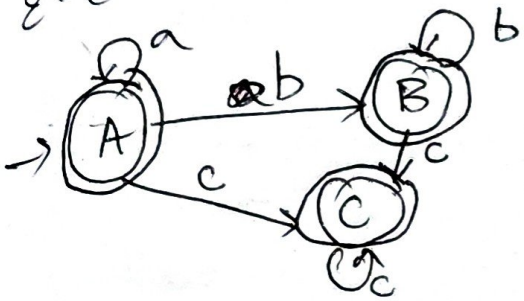
ε-closure (B, c) = ε-clos(q_2) = $\emptyset$ C

C - $\{q_2\}$

ε-closure (C, a) = $\emptyset$

ε-closure (C, b) = $\emptyset$

ε-closure (C, c) = $\{q_2\}$: C



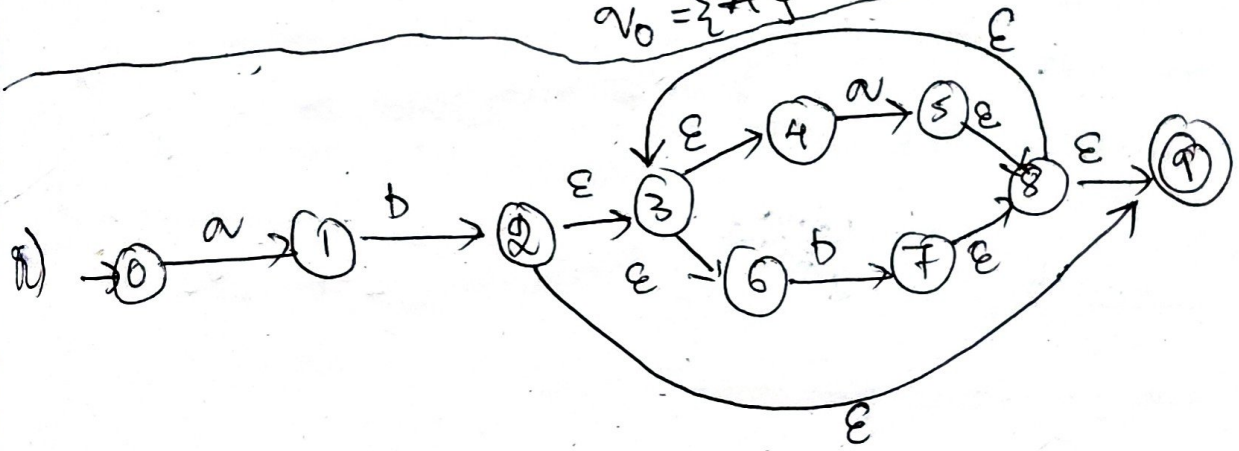
$M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{A, B, C\}$

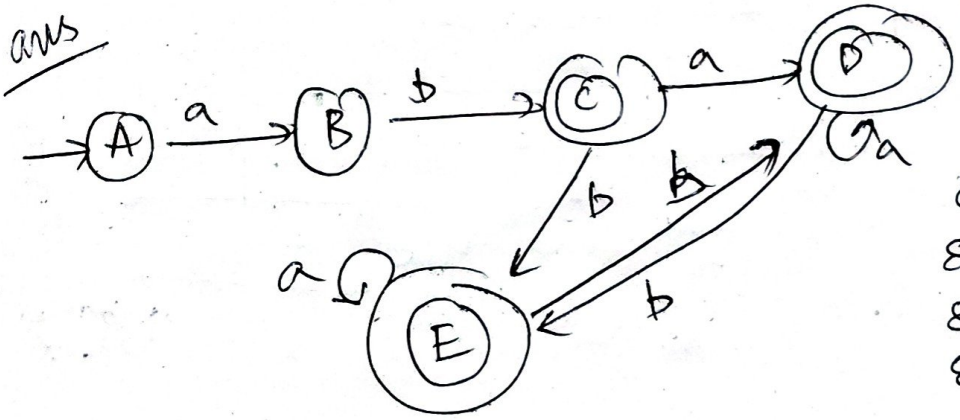
$\Sigma = \{a, b\}$ $F = \{A, B, C\}$

$q_0 = \{A\}$

ii)



ans



$\varepsilon(0) = 0$

$\varepsilon(1) = 1$

$\varepsilon(2) = \{2, 4, 3, 6, 9\}$

$\varepsilon(3) = \{3, 4, 6\}$

$\varepsilon(4) = \{4\}$

$\varepsilon(5) = \{5, 8, 9\}$

$\varepsilon(6) = \{6\}$

$\varepsilon(7) = \{7, 8, 9\}$

$\varepsilon(8) = \{8, 9, 3\}$

$\varepsilon(9) = \{9\}$

$\varepsilon(\{0\}, a) = \varepsilon(1) = 1$ - B

$\varepsilon(\{0\}, b) = \varepsilon(\emptyset) = \emptyset$

Minimization of DFA

* Two states p & q of DFA are equivalent (indistinguishable) if & only iff $\delta(p,w)$ & $\delta(q,w)$ are final state or both $\delta(p,w)$ & $\delta(q,w)$ are non-final state $\forall w \in \Sigma^*$ i.e. $\delta(p,w) \in F$ & $\delta(q,w) \in F$ where F is final

(or)

$\delta(p,w) \notin F$ & $\delta(q,w) \notin F$

* Two states p & q are distinguishable if there is atleast one string w such that one of $\delta(p,w)$ & $\delta(q,w)$ is final state & the other is non-final state then the states p & q are not Equivalent & called distinguishable state

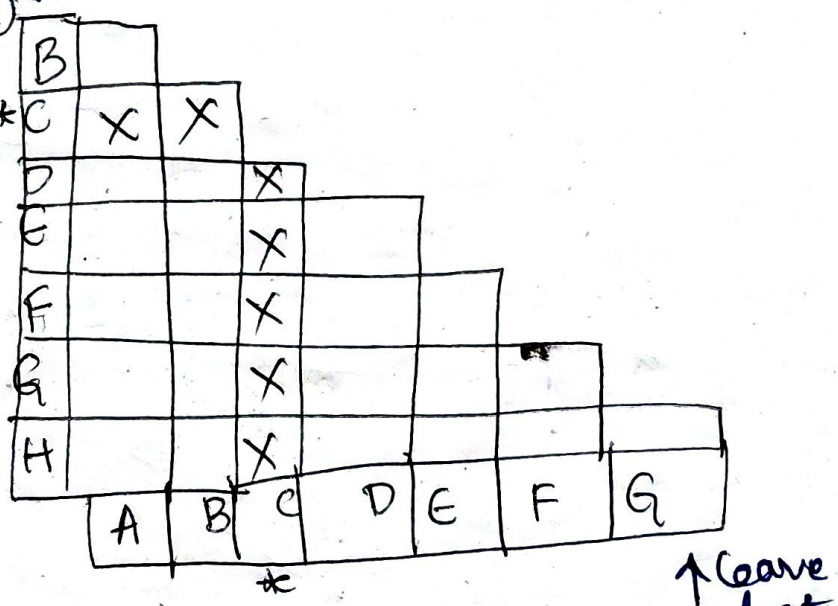
* We use Table Filling Method

(Q) Minimize the DFA using Table Filling algo or Method

δ	a	b
→A	B	F
B	G	C
*C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

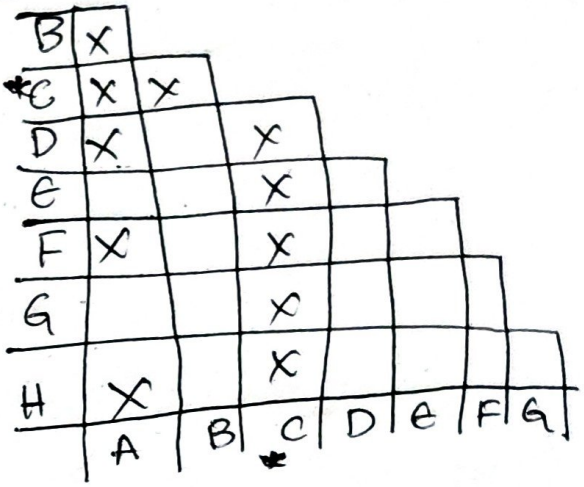
Leave first

X → distinguishable 1-F, 1-NF



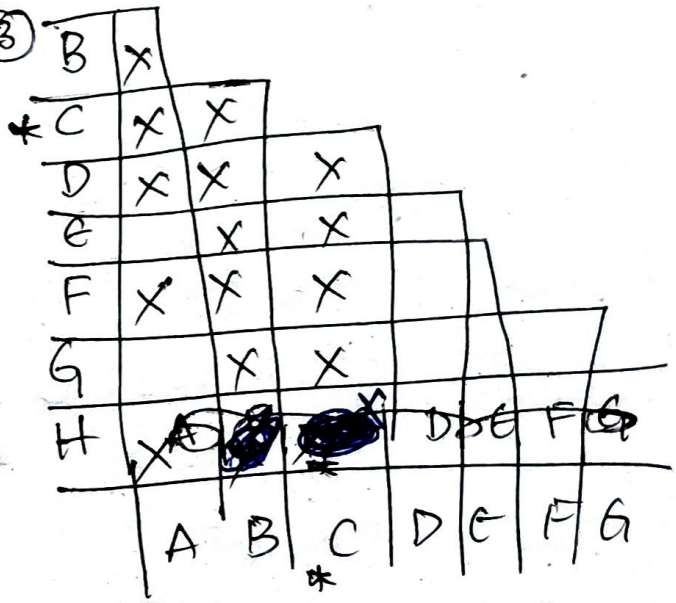
unmarked δ	a	b
(A, B)	(B, G)	(F, C)
(A, D)	(B, C)	(F, G)
(A, E)	(B, H)	(F, F)
(A, F)	(B, C)	(F, G)
(A, G)	(B, G)	(F, E)
(A, H)	(B, G)	(F, C)

(2)

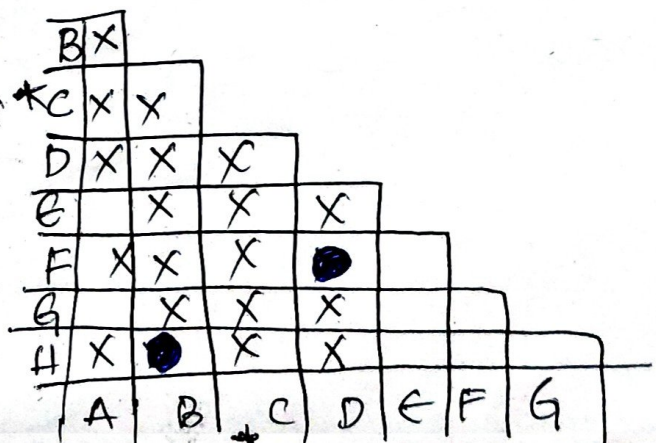


unmarked δ	a	b
(B, D)	(G, C)	(C, G) X
(B, E)	(G, H)	(C, F) X
(B, F)	(G, C)	(C, G) X
(B, G)	(G, G)	(C, E) X
(B, H)	(G, G)	(C, C)

(3)

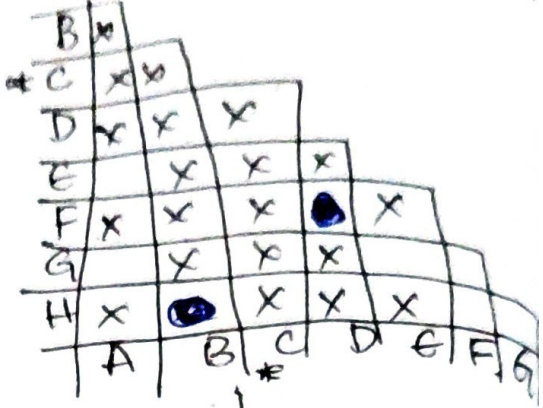


unmarked δ	a	b
(D, E)	C, H	G F
(D, F)	C, C	G G
(D, G)	C, G	G E
(D, H)	C, G	G C



unmarked E

S	a	b	
(E, F)	(H, C)	(F, G)	X
(E, G)	(H, G)	(F, E)	
(E, H)	(H, G)	(F, C)	X

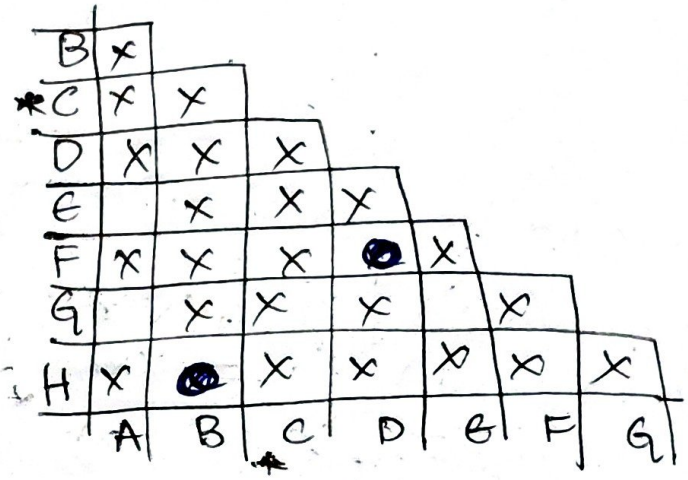


unmarked F

S	a	b	
(F, G)	C G	G E	X
(F, H)	C G	G C	X

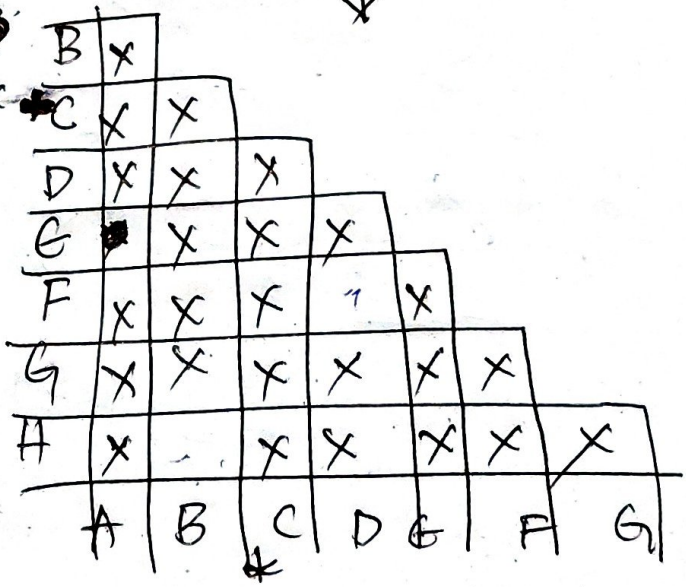
unmarked G

S	a	b	
G, H	G G	E C	X



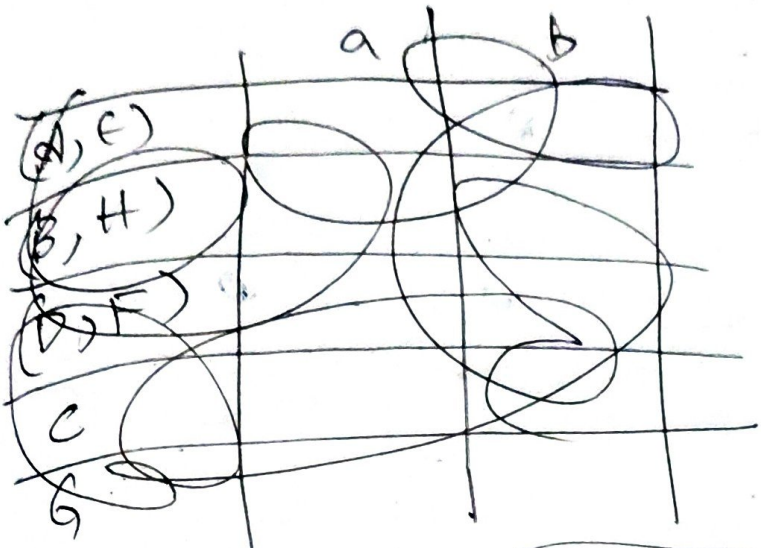
rest unmarked

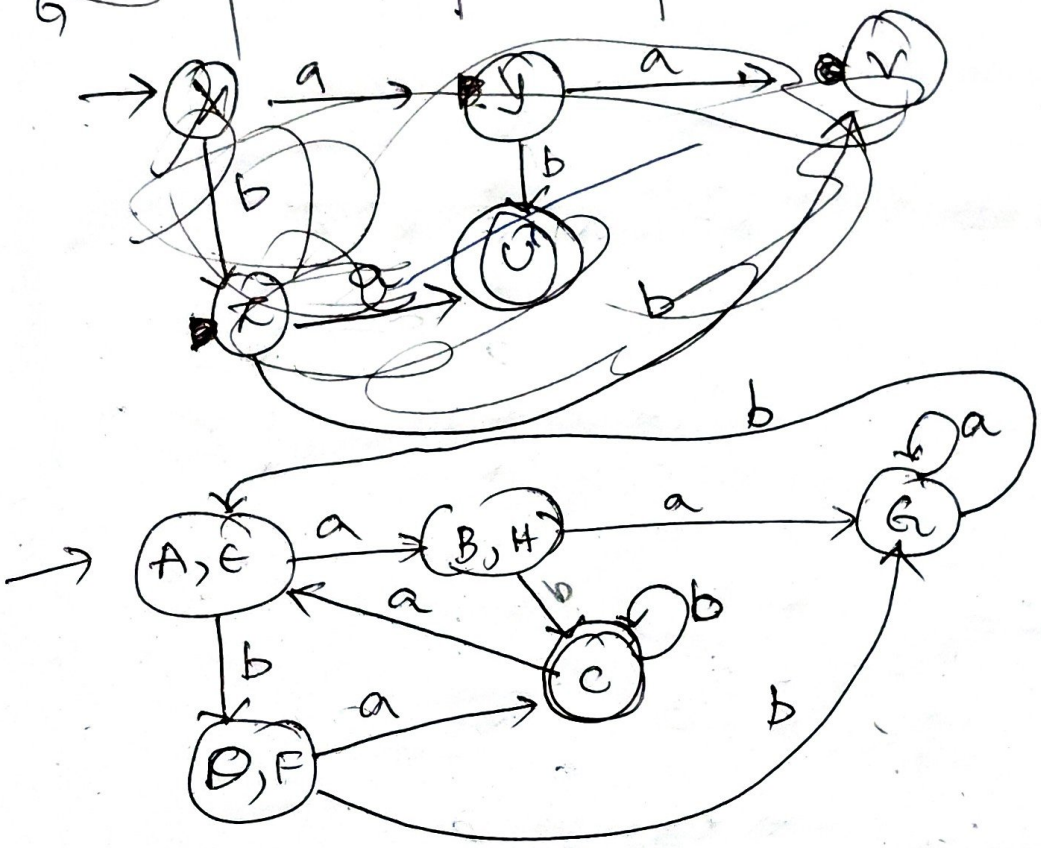
S	a	b	
(A, E)	(B H)	(F F)	
(A, G)	(B G)	(F G)	X
(E, G)	(H G)	(F F)	X
(B, H)	(G G)	(C C)	
(D, F)	C C	G G	



indistinguishable pairs

distinguishable pairs





$M = \{Q, \Sigma, \delta, q_0, F\}$

$Q = \{(A, \epsilon), (B, H), G, C, (D, F)\}$

$\Sigma = \{a, b\}$

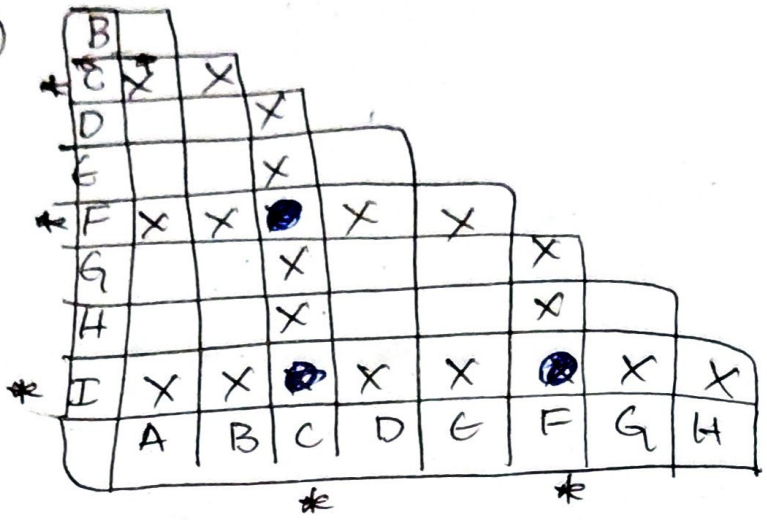
$q_0 = \{(A, \epsilon)\}$

$F = \{C\}$

Minimize DFA

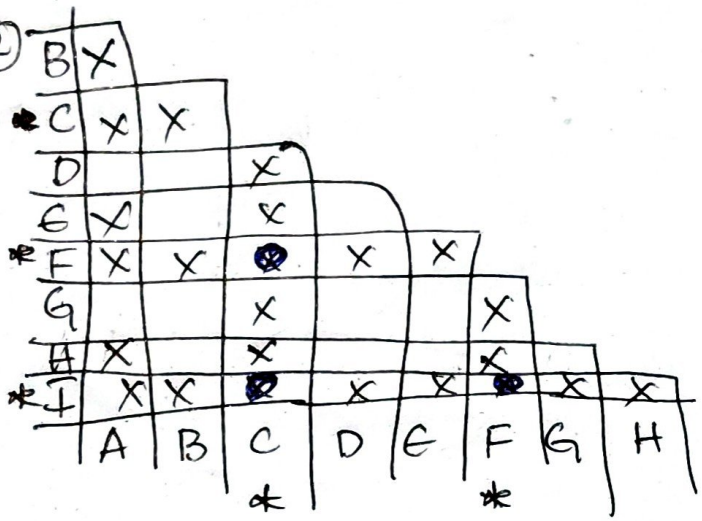
δ	0	1
→A	B	E
B	C	F
*C	D	H
D	E	H
E	F	I
*F	G	B
G	H	B
H	I	C
*I	A	E

①



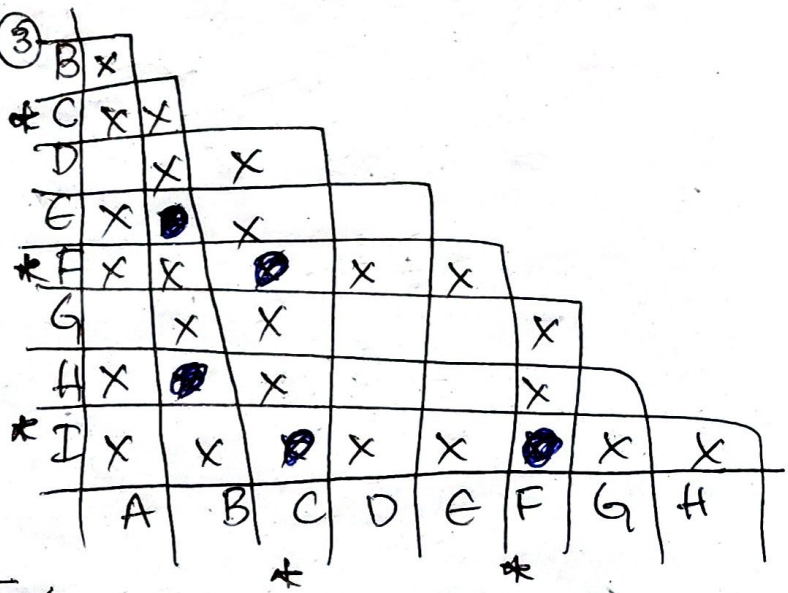
	0	1
(A,B)	(B,C)	(E,F) X
(A,D)	(B,E)	(E,H)
(A,E)	(B,F)	(E,I) X
(A,G)	(B,H)	(E,B)
(A,H)	(B,I)	(E,C) X

②



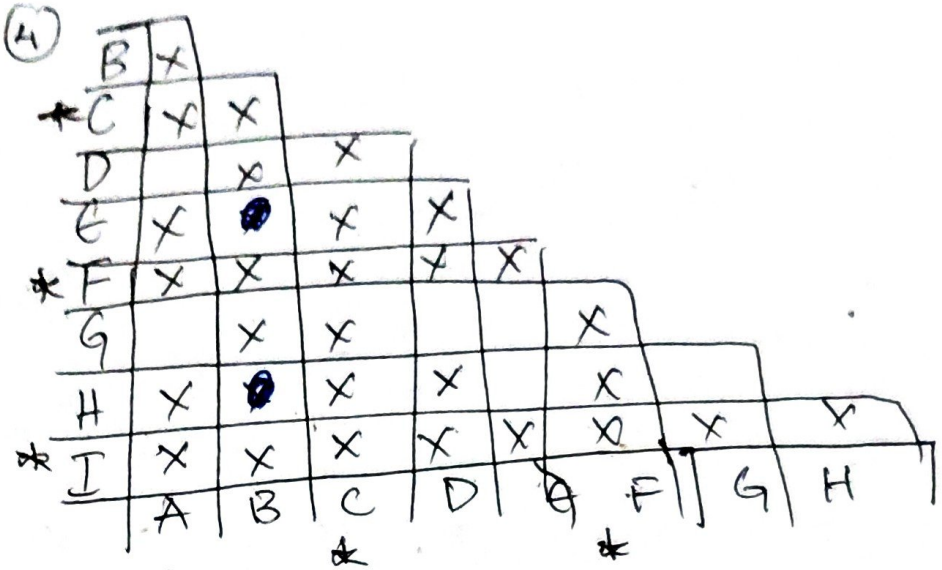
	0	1
(B,D)	C E	F H X
(B,E)	C F	F I
(B,G)	C H	F B X
(B,H)	C I	F C

③



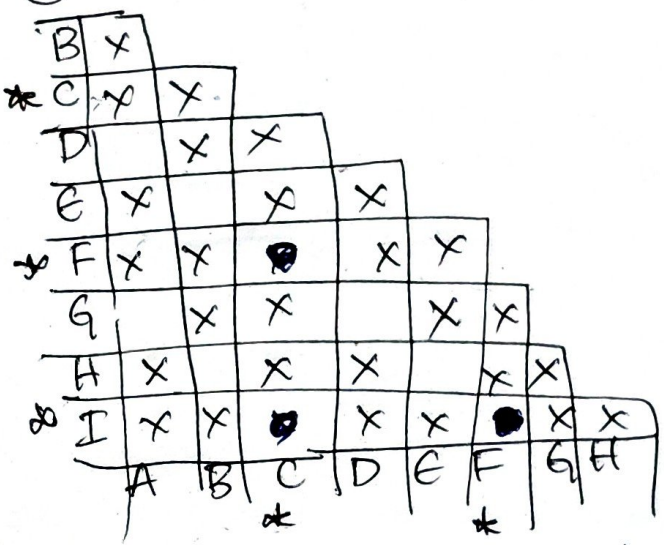
	0	1
(D,E)	E F	H I X
(D,G)	E H	H B
(D,H)	E I	H C X

④



	0	1
(E, G)	F H	I B X
(E, H)	F I	I C

⑤



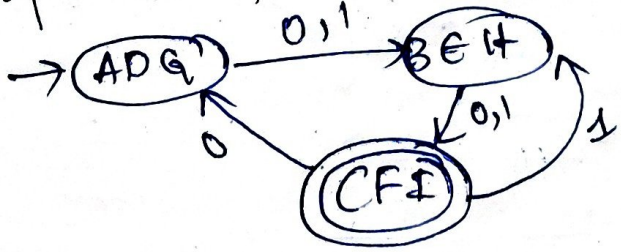
	0	1
(G, H)	H I	B C X

	0	1
(A, D)	(B, E)	(E, H)
(A, G)	(B, H)	(E, B)
(B, E)	(C, F)	(F, I)
(B, H)	(C, I)	(F, C)
(D, G)	(E, H)	(H, B)
(E, H)	(F, I)	(I, C)
(C, F)	D G	H B
(F, I)	G A	B E
(C, I)	D G	H B

similarity

	0	1
(A D, G)	B E H	E H B
(B, E, H)	C F I	F I C
	D G A	H B E

all final so (C, F, I)



Minimize DFA

S	0	1
→A	B	A
B	A	C
C	D	B
*D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

① Work this at every step

B	X						
C	X	X					
D	X	X	X				
E	X	X		X			
F	X		X	X	X		
G		X	X	X	X	X	
H	X	X	X	X	X	X	X
	A	B	C	D	E	F	G

②

	0	1	
(A,B)	B A	A C	
(A,C)	B D	A B	X
(A,E)	B D	A F	X
(A,F)	B G	A E	
(A,G)	B F	A G	
(A,H)	B G	A D	X

③

	0	1	
(B,C)	A D	C B	X
(B,E)	A D	C F	X
(B,F)	A G	C E	
(B,G)	A F	C G	
(B,H)	A G	C D	X

④

	0	1	
(C,E)	D D	B F	
(C,F)	D G	B G	X
(C,G)	D F	B G	X
(C,H)	D G	B D	X

⑤

	0	1	
(E,F)	D G	F E	X
(E,G)	D F	F G	X
(E,H)	D G	F D	X

⑥

	0	1
(F,G)	G F	E G
(F,H)	G G	E D X

⑦

	0	1	
(G,H)	F G	G D	X

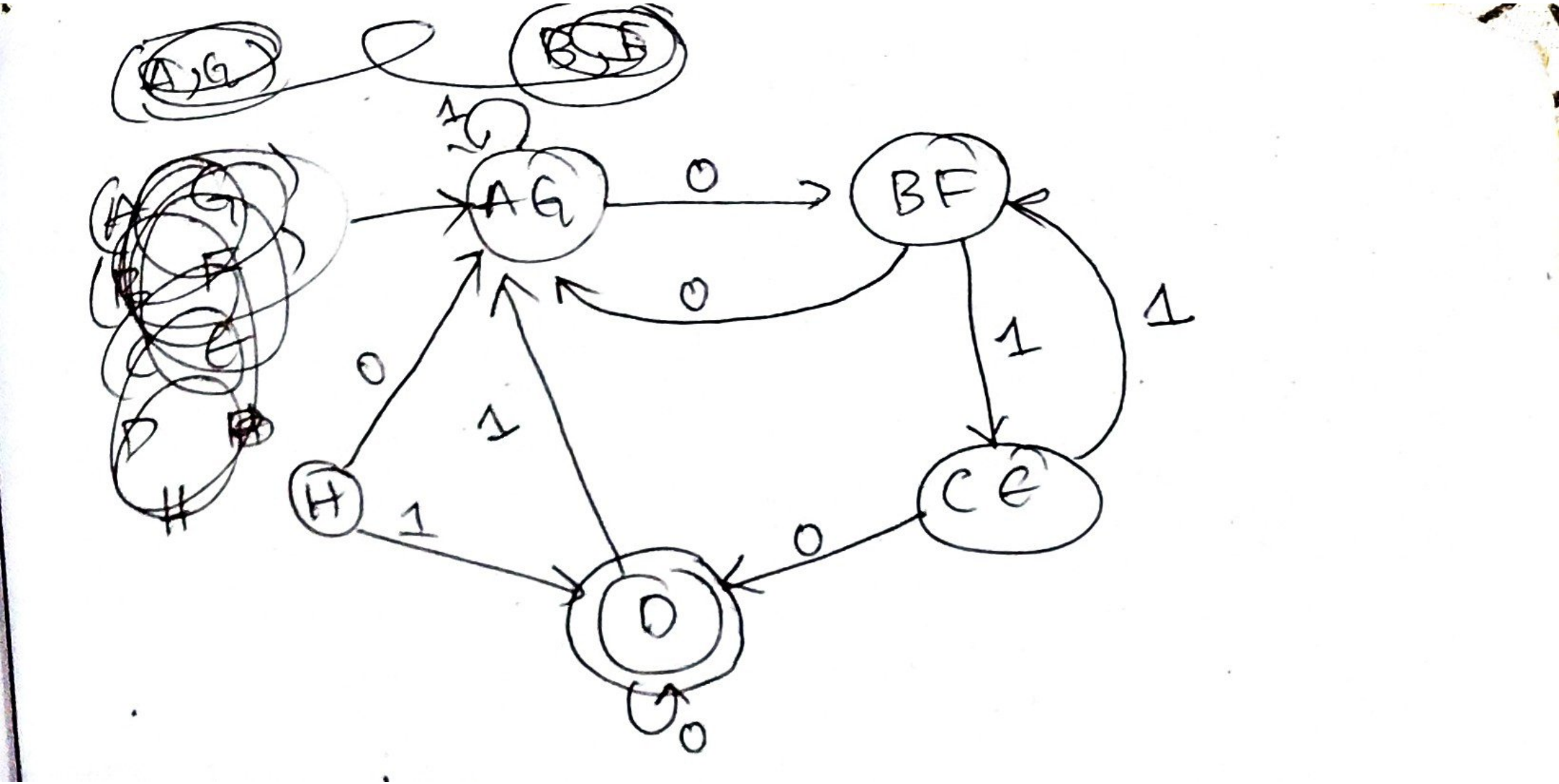
	0	1
(A,B)	B A	A C X
(A,F)	B G	A E X
(A,G)	B F	A G ✓
(B,F)	A G	C E ✓
(B,G)	A F	C G X
(C,E)	D D	B F ✓
(F,G)	G F	E G X

(A,G) (B,F)

(C,E)

(D), (H)

final


A,G
B,F
1
AG
0
BF
0
1
1
0
1
H
1
0
CE
D
0

ε-NFA problem

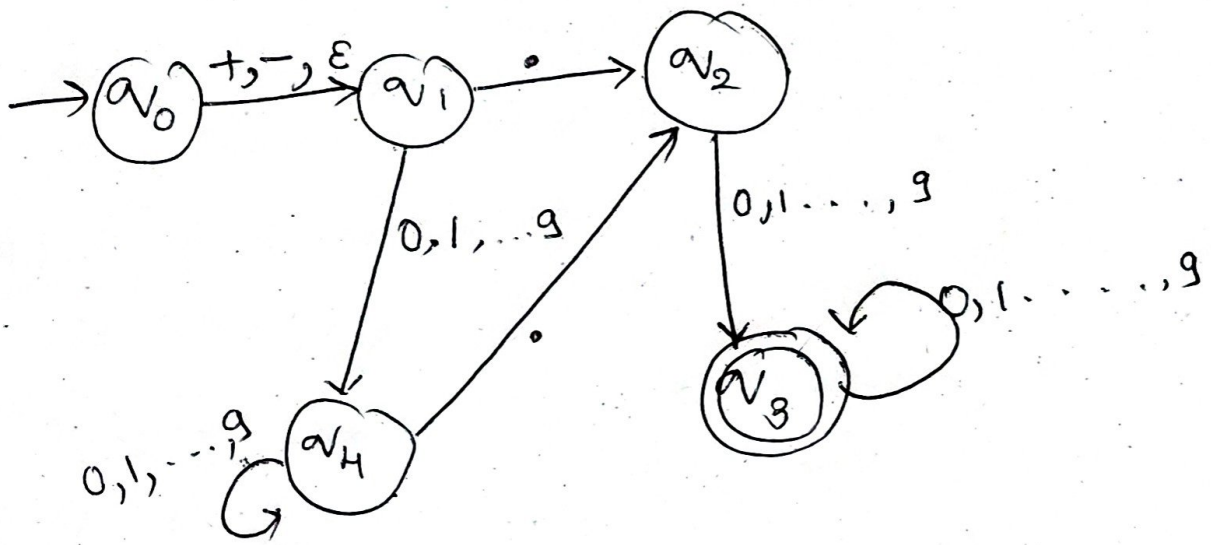
(Q) Obtain NFA with epsilon transitions to accept decimal numbers

numbers can be integers & floating point numbers

(i) an optional + or − sign followed by dot followed by string of digits

(or)

(ii) an optional + or − followed by string of digits followed by dot & in turn followed by string of digits



Now Convert to DFA

ε closure (q_0)

TOC Module 2

Regular Expression (RE)

- Regular language can be described using regular expressions consisting of symbols such as alphabets in Σ, the operators such as +, ·, * & special symbols ( ) ;
- Operator precidence (precedence)

Kleen star (*) — highest
Concatination (·)
union (+) ∪ — lowest

- Formal Defination (Regular Expre)

(i) ϕ is a regular expression denoting Empty string

(ii) ε is a regular Expression denoting Empty string

(iii) Σ is a regular Expression denoting alphabet
↳ & every Element is (RE)

(iv) if a is regular Expression which indicates the language containing only strings of a

(v) if R & S are RE then corresponding langu
L_R & L_S denotes Regular Languages
$L_R \cup L_S$ (or) $L_R . L_S$ (or) L_R^*

(vi) Expressions obtained from above all rulz again is an RE

(2) Write Regular Expression string of 0 or more a's

Ans RE = a^+

(3) atleast one a

Ans RE = a^+ or a^*a or aa^*

(4) Either one a or one b

Ans RE = $a+b$

(5) any length a's & b's including null string

Ans RE = $(a+b)^*$

(6) a's & b's Ending with abb

Ans RE = $(a+b)^*abb$

(7) a's & b's starting with ab

Ans RE = $ab(a+b)^*$

(8) a's & b's having substring aa

Ans RE = $(a+b)^*aa(a+b)^*$

(9) any a's by any b's & any c's

Ans RE = $a^*b^*c^*$

(10) at least one a followed by atleast 1 b and by atleast 1 c

Ans RE = $a^+b^+c^+$

(11) set of strings of a's & b's ending with either a or bb

Ans RE = $(a+b)^*(a+bb)$

Q. Even no of a's followed by odd no of b's

A. $RE = a^{2n} b^{2n+1} \quad n = 0,1,2 \ldots \ldots$

$RE = (aa)^* (bb)^* b$

or

$(aa)^* \; b \, (bb)^*$

Q. 0's & 1's length $<= 2$

A. $RE = (\varepsilon + 0 + 1)^2$ or $(\varepsilon + 0 + 1)(\varepsilon + 0 + 1)$

Q. Strings 0's or 1's length $<= 10$

A. $RE = (\varepsilon + 0 + 1)^{10}$

Q. Strings a's & b's having Even length

A. $RE = ((a+b)^2)^*$ multiple of 2

or $((a+b)(a+b))^*$

Q. Strings of a's & b's multiple of 3

A. $RE = ((a+b)^3)^*$

Q. strings of a's & b's having odd length

A. $RE = (a+b)^2)^* (a+b)$

Q. strings a's & b's not more than 3 a's

A. $RE = (\varepsilon + a) b^* (\varepsilon + a) b^* (\varepsilon + a) b^*$ at most

$= b^* (\varepsilon + a) b^* (\varepsilon + a) b^* (\varepsilon + a) b^*$

or

$(b^* (\varepsilon + a) b^*)^3$

(Q) Strings with 2 or more letters but beginning & ending with the same letter

Sol RE = $a(a+b)^*a + b(a+b)^*b$

(Q) a's & b's either even or multiple of 3 or both

Sol RE = $\left(((a+b)^2)^* + ((a+b)^3)^*\right)^*$

(Q) a's & b's every block of 4 consecutive symbols contains at least 2 a's

Sol RE = $(aa(a+b)(a+b) + (a+b)a(a+b)a + (a+b)(a+b)aa + (a+b)aa(a+b) + a(a+b)(a+b)a + a(a+b)(a+b)a)^*$

(Q) 0's & 1's without any consecutive 1's

Sol RE = $(\varepsilon+1)(01)^*0^*$

(Q) RE to accept a's & b's whose second symbol from the right end is a

Sol RE = $(a+b)^*a(a+b)$

(Q) a's & b's 3rd symbol from the right is a & the 4th is b

Sol RE = $(a+b)^*ba(a+b)(a+b)$

(Q) $L = a^n b^m$ / m+n is even

Sol RE = $((a+b)^2)^* + ((a+b)^2)^*(a+b)(a+b)$

$RE = (aa)^*(bb)^* + a(aa)^* b(bb)^*$

b is followed by a

(a) $\{a^n b^m \mid n \geq 1, m \geq 1, nm \geq 3\}$

sol

$m=1$	$n \geq 3$	$aaab \Rightarrow aaa\,a^*b$
$m=2$	$n \geq 2$	$aabb \Rightarrow aa\,a^*bb$
$m=3$	$n \geq 1$	$abbb \Rightarrow a\,a^*bbb$
$n=1$	$m \geq 3$	$abbb \Rightarrow a\,bbb\,b^*$
$n=2$	$m \geq 2$	$aabb \Rightarrow aabb\,b^*$
$n=3$	$m \geq 1$	$aaab \Rightarrow aaa\,bb^*$

$RE = aaa\,a^*b + aa\,a^*bb + a\,a^*bbb + abbbb^* + aabbb^* + aaabb^*$

(a) $a^n b^m \mid n \geq 4, m \leq 3$

sol

$m=0$	$n \geq 4 \Rightarrow$	$aaaa\,a^*$
$m=1$	$n \geq 4 \Rightarrow$	$aaaa\,a^*b$
$m=2$	$n \geq 4 \Rightarrow$	$aaaa\,a^*bb$
$m=3$	$n \geq 4 \Rightarrow$	$aaaa\,a^*bbb$

(a) $w \mid |w| \bmod 3 = 0$ $\quad w \in \{a,b\}^*$ → multiple of 3

sol

$RE = ((a+b)^3)^*$

(a) $w \mid n_a(w) \bmod 3 = 0$ $\quad w \in \{a,b\}^*$

sol

$RE = (a^3)^* b^* \Rightarrow (a^3 b)^*$ wrong

$RE = (b^* a b^* a b^* a b^*)^*$

(Q) do not end with 01 $w \in \{0,1\}$

Ans $RE = (0+1)^* (11 + 10 + 00)$

BUILDING an FSM (finite state machine) from a regular expression

THEOREM

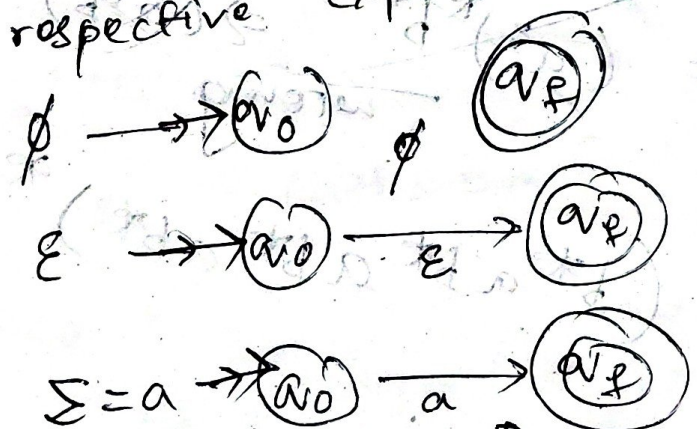
- For Every regular expression there is an equivalent FSM/FA any language that can be defined with a RE can be accepted by some finite machine & so is regular

Proof: Let R be a regular expression then there exists finite automata $M=(Q, \Sigma, \delta, q_0, F)$ which accepts the language $L(R)$

By definition
1. ϕ is RE
2. ε is RE
3. $\Sigma = a$ is RE

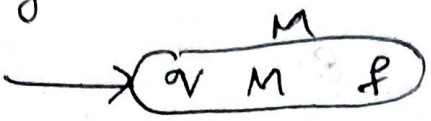
~~4. R & S operations on RE~~

The machines to recognize the language for respective expression are



We know that if R & S are ~~RE's~~ then R & S are RE then R+S, R·S, R^* are RE

Schematic representation of RE R corresponding to Language L(R) is given by

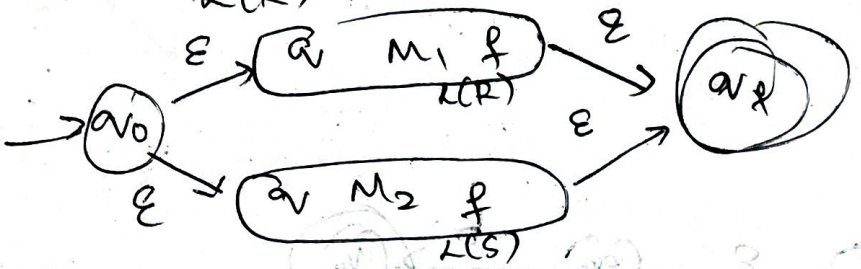


q → start
f → final

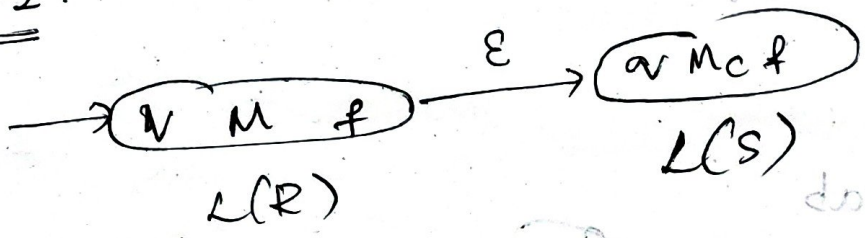
$M = (Q, \Sigma, \delta, q_0, F)$

Case 1 $R = R + S$

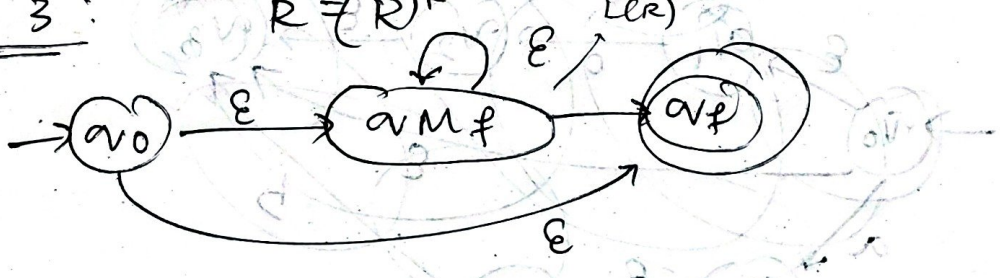
$L(R) + L(S)$



Case 2: $R = R.S$



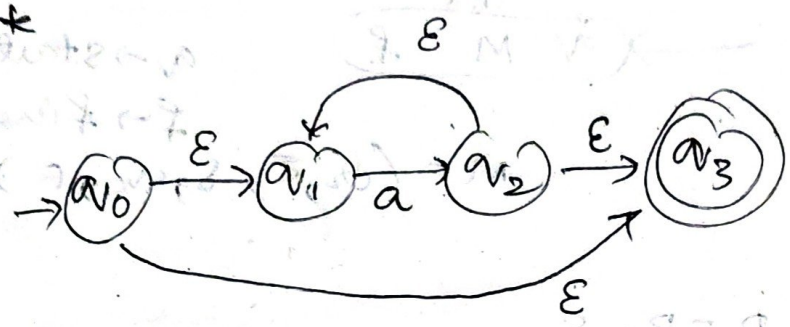
Case 3: $R = (R)^*$



Obtain Finite Automata for a RE construct

(a) a^*

Sol:

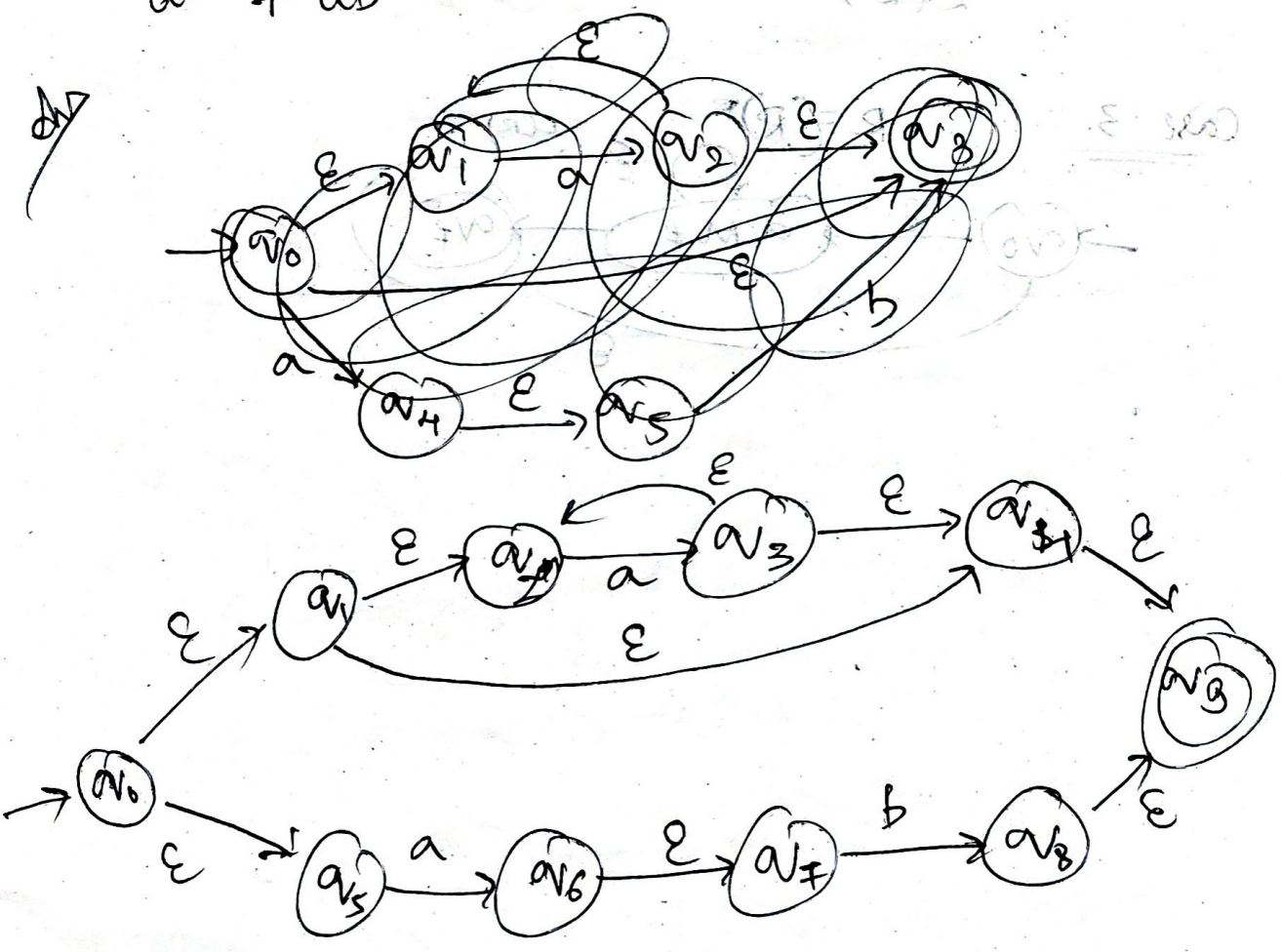


(b) ab

Sol:

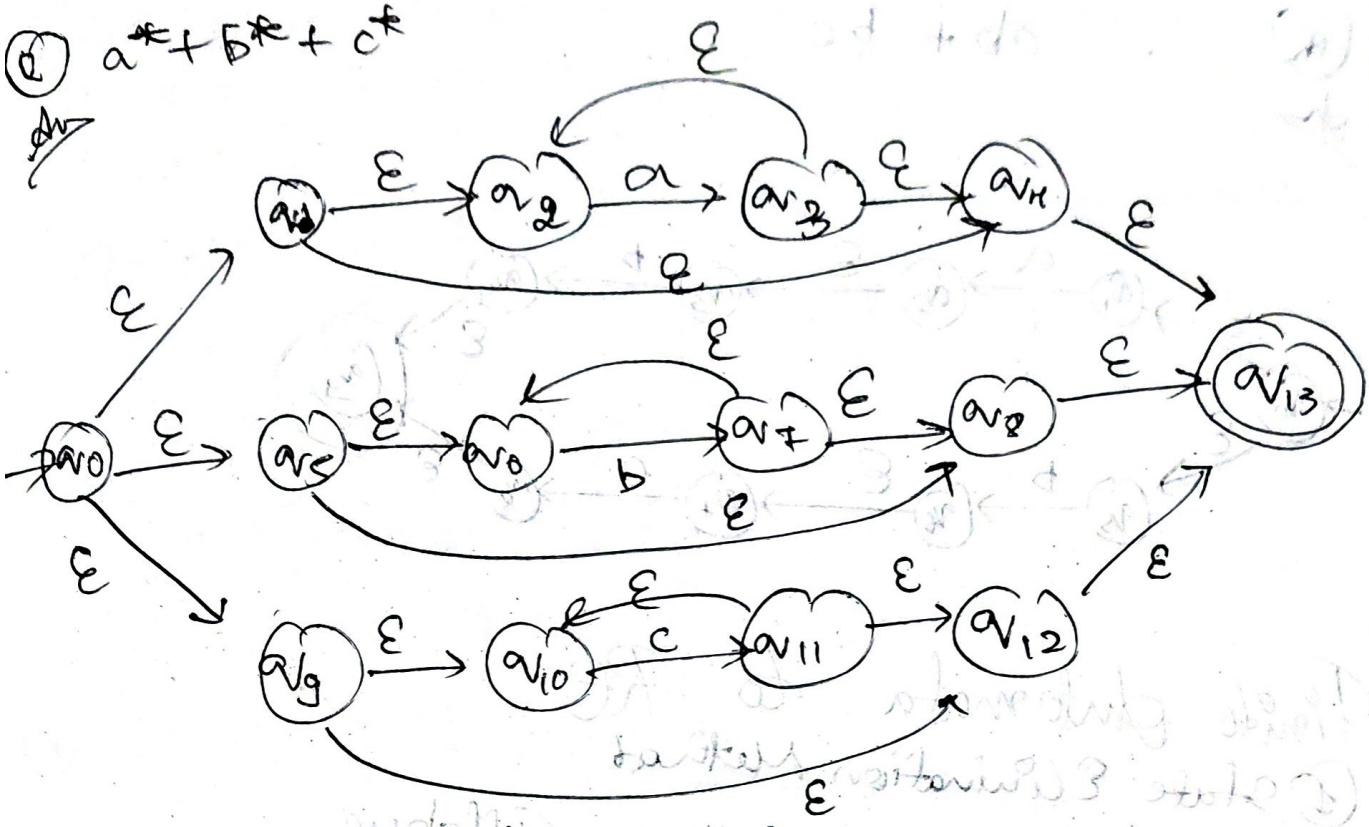
(c) $a^* + ab$

Sol:



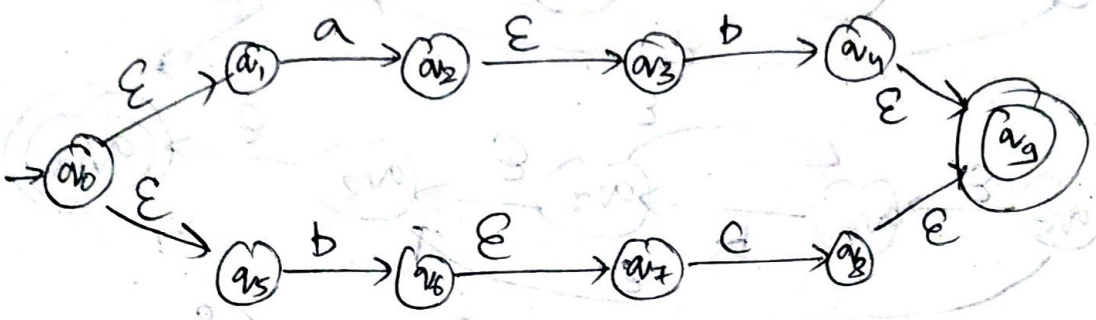
(Q) $a^* + b^* + c^*$

Ans



(Q) $(a^* + b^* + c^*)^*$

(8) $ab + bc$



Finite automata to RE

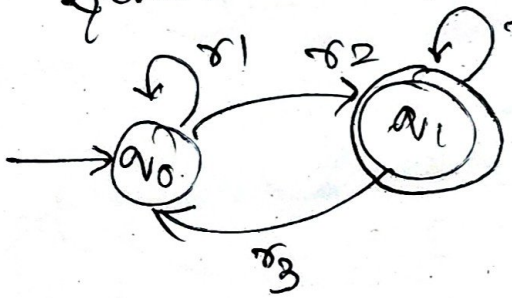
(1) State Elimination Method

(2) Kleene theorem X Not Syllabus

(1) State Elimination Method

Let $M = (Q, \Sigma, \delta, q_0, F)$ be FA Recognising the language L then there Exisits equivalent RE R for Regular Language L Such that $L = L(R)$

Consider Generalised Graph



$$r = r_1^* r_2 (r_4 + r_3 r_1^* r_2)^*$$

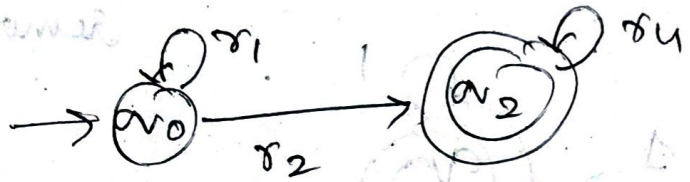
$$= r_1^* r_2 r_4^* + r_1^* r_2 (r_3 r_1^* r_2)^*$$

NOTE
① Each final state a apply the Reduction procedure bring the graph to be Generalized form

② If the start state is also accepting state.

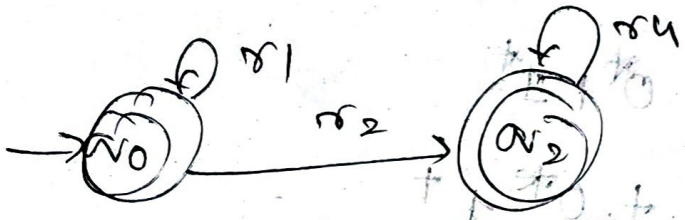
RE = r_1^*

③



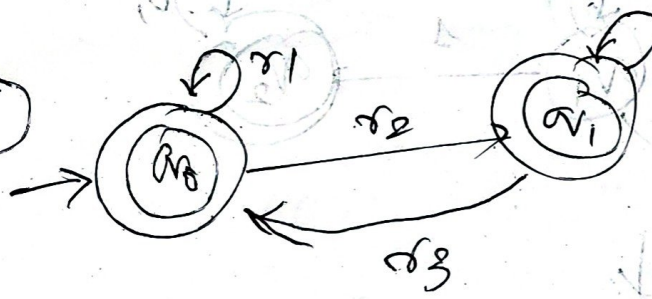
$r = r_1^* r_2 r_4^*$

④



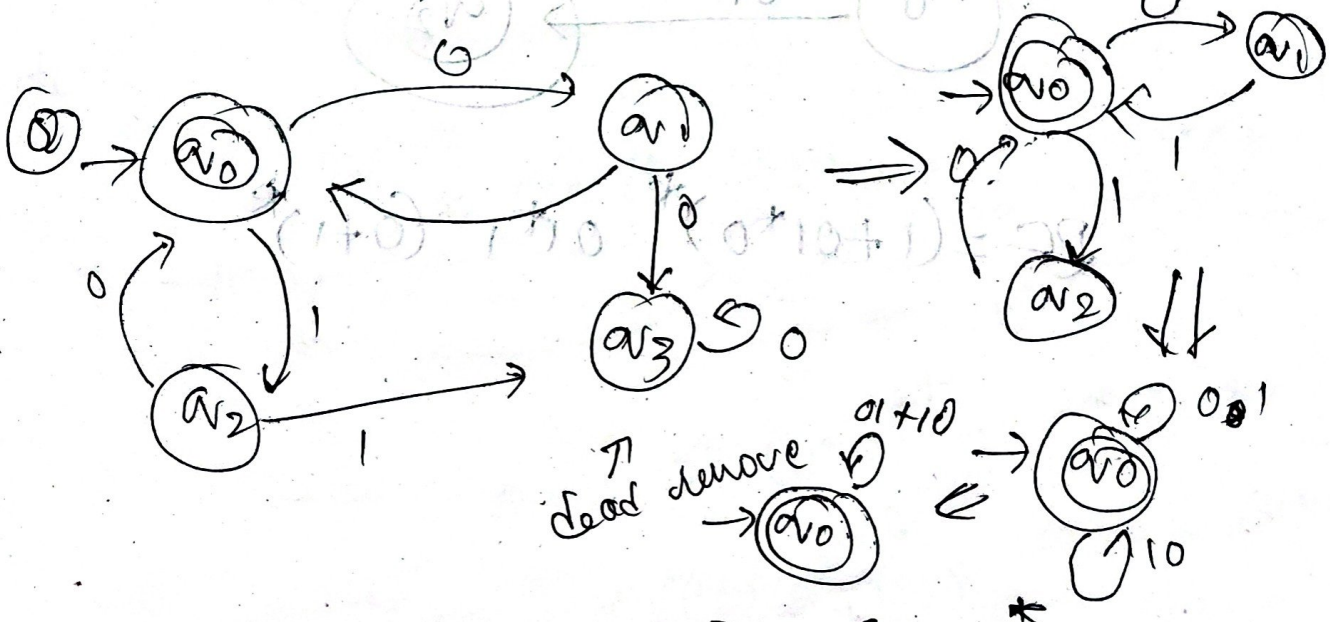
$r = r_1^* + r_1^* r_2 r_4^*$

⑤



$r = r_1^* + r_1^* r_2 (r_4^* + r_3 r_1^* r_2)^*$

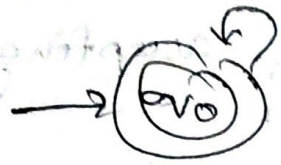
⑥



RE = $(01+10)^*$

(a)

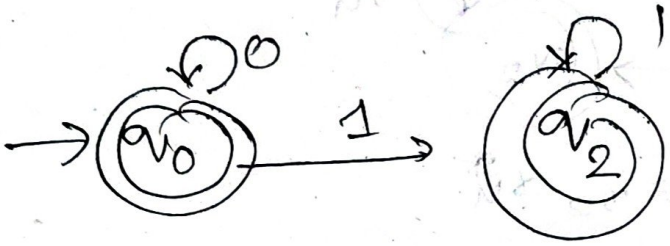
Sol



$RE = (0)^*$

q_1 & q_2 are trap directly remove

(a)

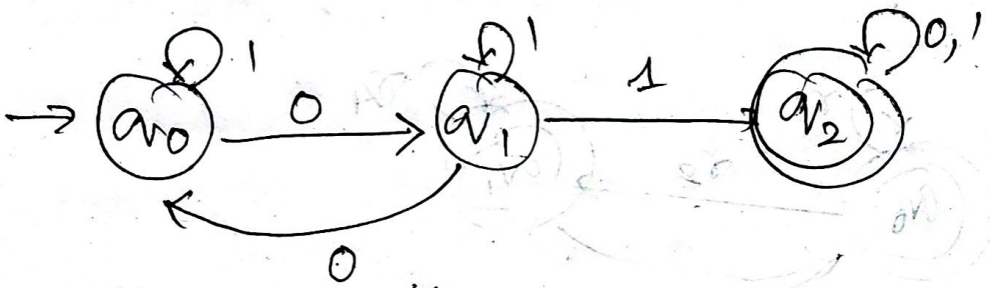


Sol

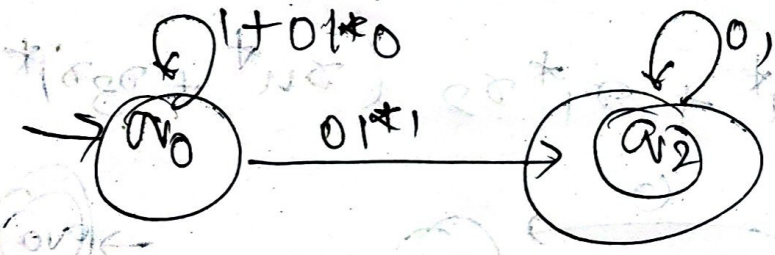
$$RE = 0^* + 0^* 1^*$$

$$= 0^* + 0^* 1^+$$

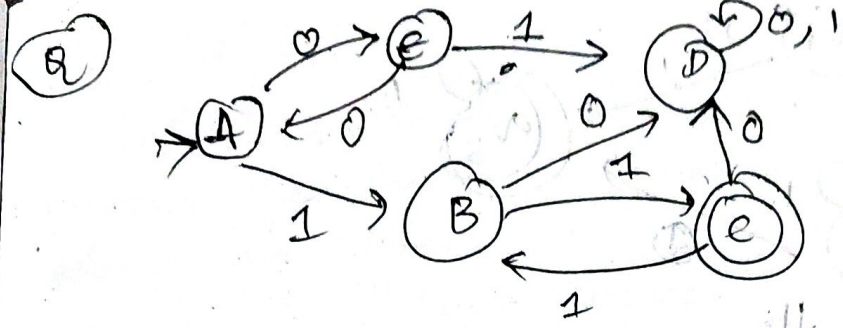
(a)

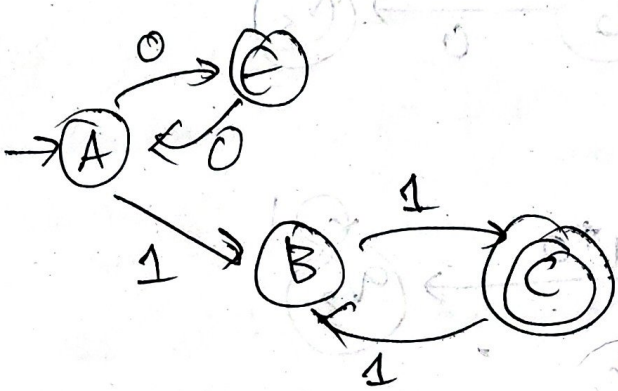


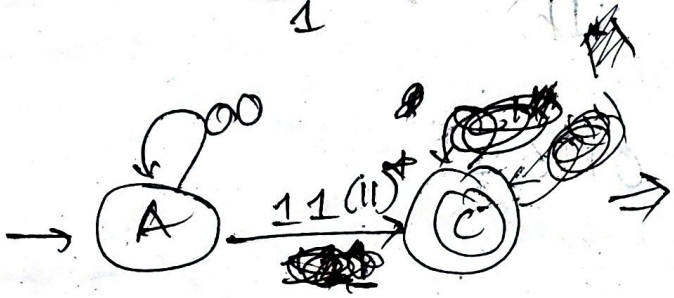
Sol



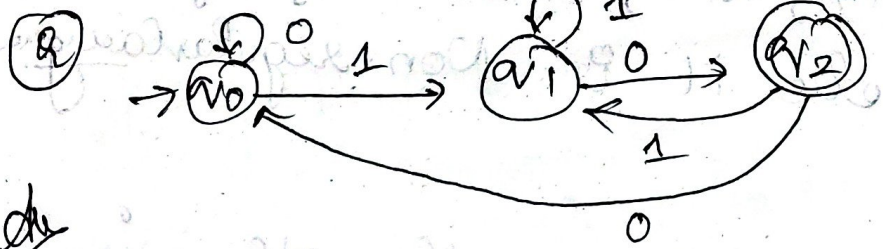
$$RE = (1+01^*0)^* \; 01^* 1 \; (0+1)^*$$

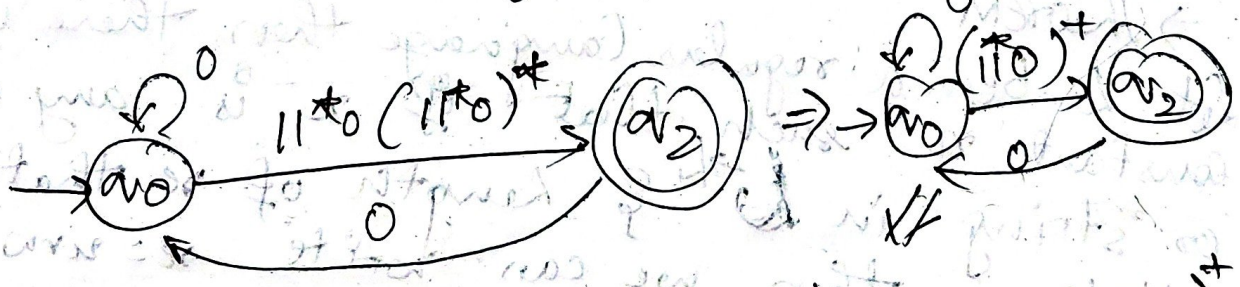






$(00)^* 11 (11)^*$

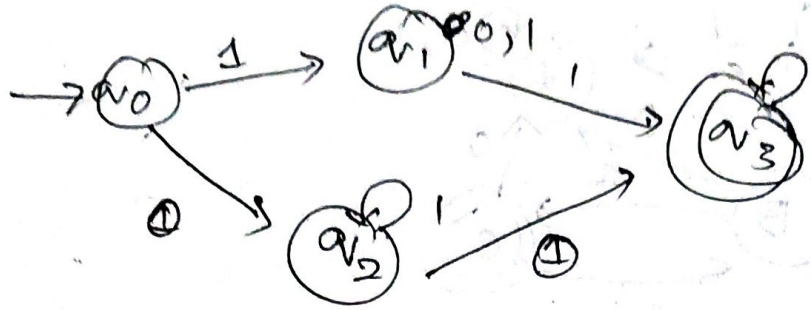


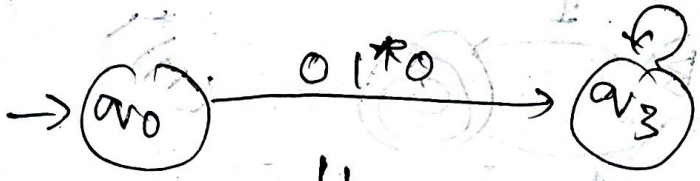


$0^* (11^*0)^+ + 0^* (11^*0)^+ 00^* (11^*0)^+$

$0^*(1+0)^+ + 0^*(1+0)^+ 0^+(1+0)^+$

(2)





01^*01^*

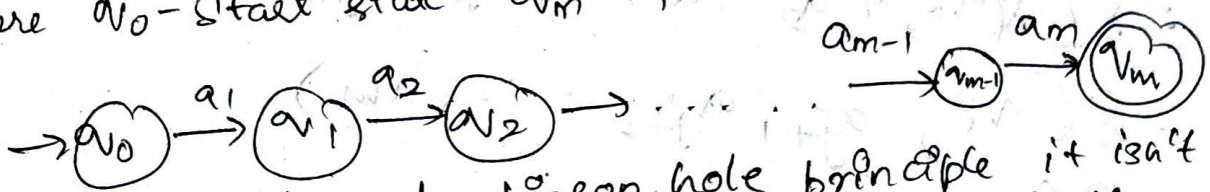
PUMPING LEMMA

To show that the language is non-regular any language for which it is possible to design FA (DFA, NFA, ε-NFA) & regular exp is RE else it is Non-regular language.

→ Theorem

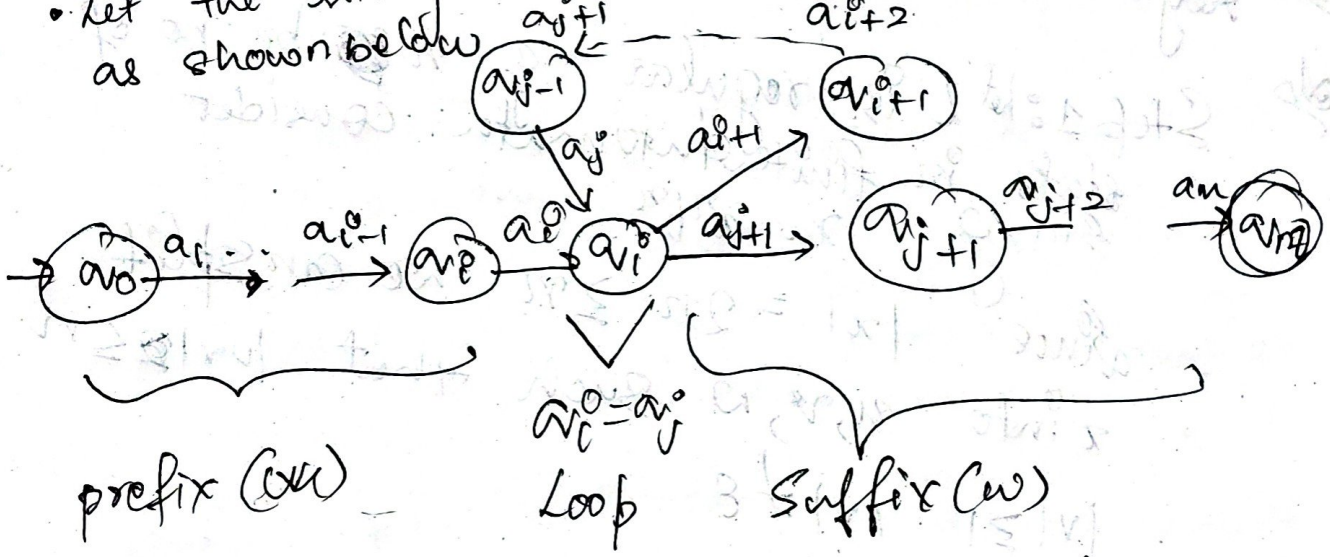
Let L be a regular language then there is constant n such that if x is any word or string in L & length of x that is $|x| \geq n$ then we can write $x = uvw$ such that $|uv| \leq n$ & $|v| \geq 1$ i.e. $v \neq \varepsilon$ & $\forall i \geq 0$ then $uv^i w \in L$ then n should not be greater than the no of states

Proof: If L is regular it is accepted by a DFA $M = (Q, \Sigma, \delta, v_0, F)$ with no of states say 'n'
consider the input $a_1, a_2 \ldots a_m$, $m \geq n$
science we have m input symbols n+1 states are needed in the sequence $v_0, v_1, \ldots, v_m$
where v_0 - Start state v_m → final state



since $|x| \geq n$ by pigeon hole principle it isn't possible to have distinct transition one of the state has to loop

- Let the string x be divided into 3 substrings as shown below



String prefix w: $a_1 \ldots a_{i-1}$: Group 1
Loop String v : $a_{i+1}, a_{i+2}, \ldots a_{j-1}$: Group 2
Suffix from w: $a_{j+1} \ldots \ldots a_m$: Group 3
if $i=0$ min string accepted by Finite automata
uv but when $i \geq 1$ the string uvvw accepted by DFA

If the string x split into substring uv^iw then if $i \geq 0$ $uv^iw \in L$ expressed using transition

$= \delta(q_0, a_1 a_2 \ldots a_{i-1} a_{i+1} \ldots a_{j+1}, a_{j+2} \ldots a_m)$

$= \delta(\delta(q_0, a_1 a_2 \ldots a_i), a_{j+1} a_{j+2} \ldots a_m)$ $m \geq 2$

$= \delta(q_i, a_{j+1} a_{j+2} \ldots a_m)$

$= \delta(q_m)$

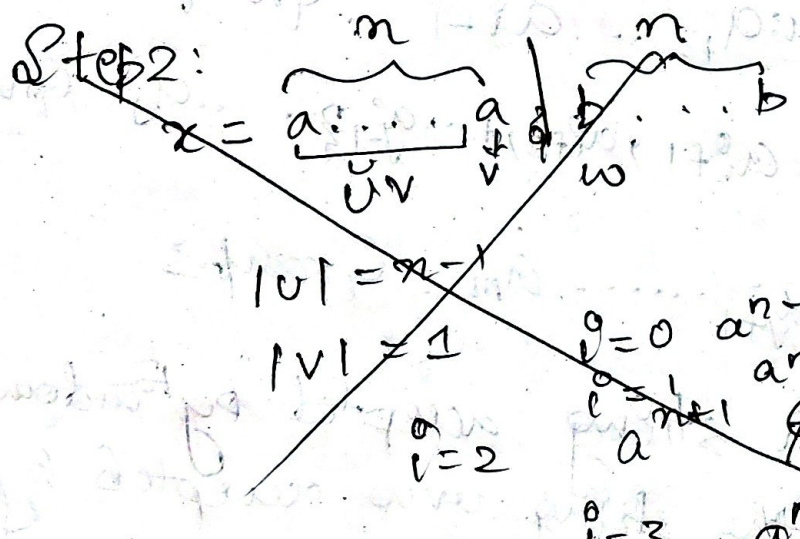
Q) Show that $L = \{a^n b^n \mid n \geq 0\}$ is not regular

Sol Step 1: let L is regular & n be no of states in finite automata consider string $x = a^n b^n$

since $|x| = 2n \geq n$ we can split x into u, v, w such that $|uv| \leq n$

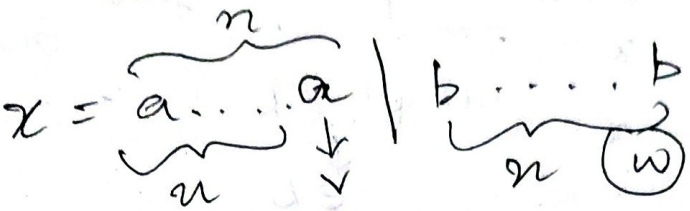
$|v| \geq 1$ $|v| \neq \varepsilon$

Step 2:



Step 3:

Step2: Since $|x| \geq 2n \geq n$ we can split



$\|uv\| \leq n$	$i=0$	$a^{n-1}b^n = x \notin L$
$\|v\| = 1$	$i=1$	$a^n b^n = x \in L$
	$i=2$	$a^{n+1}b^n = x \notin L$

$\therefore \forall\, i \geq 2 \quad x \notin L$

Step3: According to Pumping lemma $uv^iw \in L \;\forall\, i \geq 0$

if $i=0$ x contains less no of a's than no of b's

if $i \geq 2$ the string x contains more a's compared to b's

But according to pumping lemma $n_a = n_b$ (Equal a's & b's)

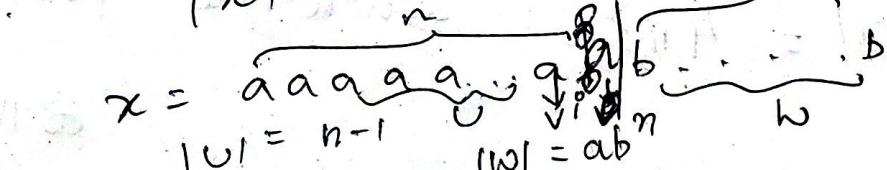
which is contradiction to the assumption that the given Language is not Regular

(Q) Show that $L = \{ a^i b^j \mid i \geq j \}$ is not regular Language.

Sol Step1: L is regular & n be no of states in Finite automata

consider string $x = a^{n+1} b^n$

Step2: $|x| = 2n+1 \geq n$ we can split uvw $i \geq 0$, $|uv| \leq n$, $|v| \geq 1$, $\exists\, |v|$



$i=0$ $\quad x = a^n b^n \notin L$ → fails so its not regular

$i=1$ $\quad x =$

$i \geq 2$ $\quad x = a^{n+2} b^n \in L$

$i=3$ $\quad \downarrow \in L$

step 3: according to pumping lemma $uv^i w \in L \forall i \geq 0$

if $i=0$ x contains equal no of a's & b's which contradicts to our assumption

hence the string is not regular

(8) $L = \{ a^n b^l c^{n+l} \mid n, l \geq 0 \}$

$\downarrow$

$n(a) = a^n$ $\quad n(b) = a$

$\downarrow$

$a^n a^l c^{n+l}$

$\downarrow$

$a^{n+l} c^{n+l}$

$\downarrow$

$L = a^n b^n$

(8) $L = \{ ww^R \mid w \in (0+1)^* \}$

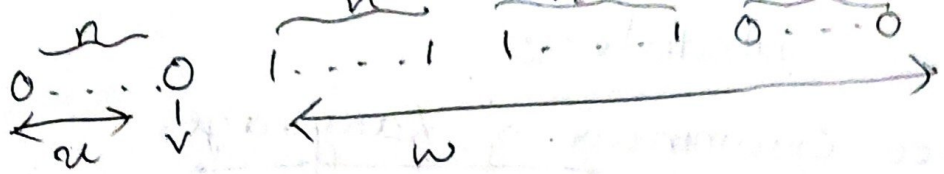
step 1:

$x = \underbrace{0^n 1^n}_{w} \underbrace{1^n 0^n}_{w^R}$

$|x| = 4n \geq n$ such that $|uv| \leq n$

$|v| \geq 1$

$\exists |v| \notin \varepsilon$



Q) Show that $L = \{ a^{n!} \mid n \geq 0 \}$ is not regular

Sol

$x = a^{n!} \quad |x| \geq n$

$x = a^j a^k a^{n!-j-k}$

$u \; v \; w$

$i = 0$

Q) Show that $L = a^n \mid n$ is prime.

Sol

Properties of Regular Language

1. Union of two Regular Language is regular
2. Complement of RL is R
3. Intersection — " —
4. Difference — " —
5. Closure operation — " — is closure
6. Concatination of RL is R
7. Reverse of RL is R
8.